History-dependent gravity and cosmological release

Galaxy, cluster, and cosmological tests

Jacob Chinitz

October 24, 2026

Abstract

Based on the Entropic Scalar EFT manuscript, its Lean proof package, the archived galaxy calculations, and the tied-abundance cosmological release calculations. The comparisons use specified source histories, transport models, and observational likelihoods; their mathematical premises and numerical domains are stated with each result.

Fifteen test families, with their evidence and open limits, are organized as distinctive predictions and discriminating tests, and recovery of established phenomenology. The same retained stock settles onto the galactic response law through contacts biased by the response cell's Bose statistics. Outside galaxy supports, cluster stock has no active-carrier cells and retains the cohort ratio.

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1 The empirical picture

A common microscopic construction ties the galactic acceleration scale, the initial gravitating stock, and its conversion into vacuum energy. The tests below examine those connections across different formation histories and established gravitational and cosmological observations.

Distinctive tests compare common coefficients and inherited stock with specified alternatives. LMC masses, Crater II stripping, and NGC 628 vertical gravity give favorable results against the tested QUMOND prescriptions or NFW populations. High-redshift kinematics remains a tension. Established benchmarks assess recovery of acceleration, lensing, acoustic spectra, expansion distances, cluster gravity, and tensor propagation. Their evidential status is stated separately for each test.

Table 1: All fifteen test families in the evidence register. Placement identifies the role of a test; the status column states its present reach. The register is exploratory, with the existing outcomes known.

ID	Test	Present result and limit	See
I. Distinctive predictions and discriminating tests			
S01	High-redshift kinematics	Reference line projection agrees; galaxy catalogue residuals retain a high-redshift tension.	3
S02	CMB–galaxy normalization	Exponential-fit $a_0 = 1.25883 \times 10^{-10}\,\mathrm{m\,s^{-2}}$; standard linear-depletion value $1.25005 \times 10^{-10}\,\mathrm{m\,s^{-2}}$; conditional mass and survival factors.	6.1
S10	Binaries and Solar System	Cassini null benchmark is compatible; a Gaia catalogue likelihood remains open.	6.5
S11	G and lepton masses	Close selected-formula values; finite routing tests leave discovery bias unquantified.	6.2
S13	Milky Way, LMC, streams, tilt	LMC mass agreement and repeated tilt realizations; Galactic mass and a joint stream fit remain unresolved.	4, 6.3
S14	Satellites and tidal dwarfs	Crater II favors stock within the tested grids; Antlia 2 and odds depend strongly on model error.	5
S15	Old-disk vertical gravity	NGC 628 gives $\chi^2 = 2.04$ for five dispersions; a broader tracer and force comparison is needed.	6.4
II. Recovery of established phenomenology			
S03	Local RAR and BTFR	Broad SPARC diagnostic and small BTFR scatter; the score is sensitive to velocity errors.	7.1
S04	Galaxy weak lensing	Updated fifteen-bin diagonal score 10.06; reduction sensitivity and joint covariance remain.	7.2
S05	CMB acoustic peaks	Planck-based matched fit and ACT comparison recover similar quality to the reference cosmology.	7.3
S06	Growth and Weyl power	ACT lensing score 16.21 over ten bins; DESI full-shape and nonlinear growth tests remain open.	7.3
S07	Dark-energy history and distances	Matched BAO/SN fit with exponential allocation; the feedback-resolved retention occupation gives $\Omega_\Lambda = 0.709 \pm 0.02$ against 0.707 with no input adjusted to it, and the frozen history fits with four parameters at $\Delta\chi^2 = +1.07$.	8.6
S08	Relaxed and colliding clusters	Positive stock targets and conserved encounters; outlier mechanisms and lens-map fits remain conditional.	9
S09	Slip and E_G	Transfer predictions are compatible with two published estimates; a joint window-convolved fit is needed.	7.4
S12	Waves and Lorentz tests	Luminal tensor branch is compatible with the polarization benchmark; operator-level Lorentz tests remain open.	7.5

[U1; CL1–CL3; D1–D2; T6]

The following calculations retain their individual likelihoods, selection effects, and error models. A simulation realization, a fitted relation, a conditional normalization, and an observation predicted before measurement have different evidential roles. Lean checks the stated mathematical implications; observed agreement is assessed separately for each test.

Relation to the theory and CDT accounts. The effective astrophysical calculations use *Entropic Scalar EFT*. The companion CDT account distinguishes the electron-anchored scale G_*, the selected bare coefficient $G_0 \simeq 1.03002 G_*$, and an interacting response G_R. The measured Newton constant remains the effective input here. Matter response and the continuum phase of the $\alpha = 7$ CDT branch retain their own tests. [T1–T5]

2 The common microscopic coefficients

2.1 Capacity, matter, and an observable gravitational field

The starting hypothesis assigns finite entanglement capacity to a tetrahedral substrate. Matter consists of persistent defects that commit some of this capacity. Mass measures the commitment in energy units. The ordinary gravitational branch identifies a capacity deficit with the weak-field metric potential. In the manuscript's convention,

$$\frac{\delta S}{S_\infty} = -\frac{2\Phi}{c^2}. \tag{1}$$

For a negative Newtonian potential, the corresponding capacity deficit is positive. The ordinary metric branch uses the measured Newton constant. Varying the specified two-potential action independently in Φ and Ψ gives

$$\nabla^2 \Psi = 4\pi G\rho, \qquad \nabla^2(\Phi - \Psi) = 0. \tag{2}$$

For smooth isolated sources with $\Phi - \Psi \to 0$ at infinity, the maximum principle gives $\Phi = \Psi$. These steps have kernel-checked proofs, connecting the reduced action to the Newtonian force and the leading no-slip metric. The physical selection of that action and its normalization retain their stated premises. [T4]

A stellar velocity, a stream deflection, and a gravitational lens measure the field generated by the effective stress-energy. They do not directly identify the microscopic material carrying that stress. The galaxy runs represent retained capacity by collisionless particles. Those numerical particles sample a positive mass distribution; the theory interprets their mass as substrate commitment. Thus the model contains an additional gravitating component even though it proposes no new species of dark-matter particle. Its physical origin and relation to baryons are the claims under test. [T1, sections 1, 15, 20 and O.3]

2.2 Why the selected face has seven transmitted and nine retained states

Each side of a shared face carries a primitive spin-3/2 representation, which has four states. Its paired space has dimension sixteen. Angular momentum addition gives

$$V_{3/2} \otimes V_{3/2} = V_0 \oplus V_1 \oplus V_2 \oplus V_3, \qquad 16 = 1 + 3 + 5 + 7. \tag{3}$$

The selected positive matching operator transmits the maximal-spin sector V_3. Its orthogonal complement has dimensions $1 + 3 + 5 = 9$ and supplies the retained marked register. The present/history response has the same representation content, $V_1 \otimes V_1 = V_0 \oplus V_1 \oplus V_2$. An explicit equivariant map identifies the marked register with the retained complement. Its uniqueness is

proved up to one scalar factor on each spin block; equivariant unitary freedom consists of block phases. The result fixes the allowed identification within the selected vertex. [T4]

The numbers describe dimensions of state spaces. They do not say that seven sixteenths of the physical energy passes and nine sixteenths becomes dark energy. An energy fraction requires an energy operator and an identified state. The release fraction derived below is about three percent, not 9/16.

There are four distinguishable face ports. Single-copy occupancy makes their seven-letter assignments injective. Including the two cell orientations gives

$$N_{\text{states}} = 2P(7,4) = 2\,(7 \cdot 6 \cdot 5 \cdot 4) = 1680. \tag{4}$$

Uniform weights would give entropy $\ln 1680 = 7.4265490724$. The theory weights these states by their closure defect, producing a slightly smaller entropy. [T1, sections 5–6 and B.2]

2.3 The finite sum that fixes the entropy

Let $m_i \in \{-3,-2,-1,0,1,2,3\}$ be the four distinct face labels. The tetrahedral face normals satisfy $\widehat{n}_i \cdot \widehat{n}_j = -1/3$ for $i \neq j$. On the product-state boundary realization the expectation of the squared closure operator is

$$K^2 = 48 - \frac{1}{3}\left[\left(\sum_i m_i\right)^2 - \sum_i m_i^2\right]. \tag{5}$$

The constant $48 = 4j(j+1)$ uses the transmitted spin $j = 3$. The Gibbs weights and their partition sum are

$$p_\eta(b) = \frac{e^{-\eta K^2(b)}}{Z(\eta)}, \qquad Z(\eta) = \sum_{b=1}^{1680} e^{-\eta K^2(b)}. \tag{6}$$

Three closure components supply the determinant factor $\eta^{3/2}$. Stationarity of the normalized closure evidence gives

$$0 = \frac{d}{d\eta}\left[\ln Z + \frac{3}{2}\ln\eta\right] = -\langle K^2\rangle_\eta + \frac{3}{2\eta}. \tag{7}$$

Direct enumeration and a one-dimensional root solve yield

$$\eta_* = 0.0298668443935, \qquad \langle K^2\rangle_* = 50.2229154254, \tag{8}$$

$$g = -\sum_b p_*(b)\ln p_*(b) = 7.41980002357. \tag{9}$$

Finite enumeration and the closure root have kernel-checked proofs and certified numerical bounds. Independent floating-point evaluation supplies the additional displayed digits; the verification records distinguish the two forms of calculation. [T4] The word "fixed" means fixed after the ensemble, matching rule, closure invariant, and evidence measure have been specified. It does not claim that finite capacity uniquely determines that microscopic model.

2.4 Two coefficients inherited by the astrophysical branches

The transverse galactic branch takes its response cell to be the tilt mode of the transverse frame, with inertia lever $\ell = g\,c/H$. The horizon temperature $\hbar H/2\pi$ then gives the dimensionless coefficient

$$\epsilon = \frac{g}{4\pi^2} = 0.1879457302, \qquad \epsilon^{-1} = 5.320684854. \tag{10}$$

and the acceleration scale

$$a_0(z) = \epsilon c H(z). \tag{11}$$

At $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $a_0(0) = 1.23072924 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. The standard linear-depletion fit has $H_0 = 68.458\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $a_0(0) = 1.25005 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. The exponential-allocation fit with massless neutrinos, linear lensing and a low-redshift supernova offset has $H_0 = 68.93885586416\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $a_0(0) = 1.25883 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. The galaxy calculations specify the reference or fitted normalization used in each comparison. This coefficient is obtained by evaluating a fixed prescription. The tilt-mode identification, its horizon thermalization, and the lever are the physical matching assumptions of that prescription; the lever coefficient follows from the register enthalpy and a one-nat oriented record read at the Unruh temperature of the projected field, given that reading. [T1, section 15]

In the cosmological branch, the committed state has mean energy $\varepsilon_K \langle K^2 \rangle_*$, while the three normalized record coordinates carry mean energy $3\varepsilon_\xi/2$. Their ratio is

$$\eta = \frac{\varepsilon_\xi}{\varepsilon_K} \frac{3}{2\langle K^2 \rangle_*}. \tag{12}$$

The common-conversion choice $\varepsilon_\xi = \varepsilon_K$ makes $\eta = \eta_*$. This identification also uses the manuscript's post-refresh energy accounting. The exact closure identity and the physical equality of conversion scales have different logical roles. [T1, O.4 and O.13]

Here g is the sharing entropy denoted g_{share} in the CDT account and $g_{\mathrm{share,eff}}$ in Astra. The galactic ϵ is distinct from the electron rest gap $\epsilon_e \simeq 4.18546 \times 10^{-23}$.

The same pair (g, η_*) therefore enters the galactic scale, the initial commitment budget, and the release fraction. The observational calculations test those connections together with the effective dynamics that takes them from a cell to a galaxy or an expanding universe.

Part I. Distinctive predictions and discriminating tests

3 A cosmological acceleration scale: prediction and tension

The model predicts an acceleration scale proportional to the cosmic expansion rate, with its coefficient fixed by equation (10). Using the background specified for this comparison gives

$$\frac{a_0(z)}{10^{-10}\,\mathrm{m\,s^{-2}}} = 1.23072924\sqrt{0.315(1+z)^3+0.685}. \tag{13}$$

Its linear projection over $0.33 \le z \le 1.44$ is

$$\frac{a_0^{\mathrm{line}}(z)}{10^{-10}\,\mathrm{m\,s^{-2}}} = 1.002784 + 1.225645z. \tag{14}$$

The published MOND-based kinematic analysis in Appendix E of Ciocan et al. gives an intercept 1.03 ± 0.05 and slope 1.20 ± 0.10 in these units. The main-text dark-halo decomposition gives a larger slope, about 1.59. These are distinct inferences from the kinematic analysis and must be compared separately. [1]

3.1 What is measured

For a nearly circular stellar or gas orbit, the inferred centripetal acceleration is

$$g_{\mathrm{obs}}(r) = \frac{v_c^2(r)}{r}. \tag{15}$$

A baryonic model supplies the acceleration g_b generated by the observed stars and gas. The radial acceleration relation, or RAR, describes how g_{obs} depends on g_b. In the galaxy outskirts g_b is small and the inferred attraction exceeds the Newtonian baryonic value. The parameter a_0 locates the transition.

At intermediate redshift, the rotation field, pressure support, inclination, gas profile, and stellar mass must be inferred together. Ciocan et al. explicitly describe the resulting RAR as model dependent. Their Appendix E refits the kinematics within a MOND framework and then estimates the redshift dependence. The coefficient comparison here uses that inference; it does not treat the measurements as a direct model-independent measurement of the substrate. [1]

3.2 Why a cosmological acceleration enters

The Unruh temperature associated with proper acceleration a is

$$k_B T_U = \frac{\hbar a}{2\pi c}. \tag{16}$$

For a de Sitter horizon, the corresponding temperature is

$$k_B T_H = \frac{\hbar H}{2\pi}. \tag{17}$$

Equating these two scales gives $a = cH$. This relation motivates a cosmological acceleration scale. Applying a horizon temperature to an evolving cosmological background requires the chosen horizon-contact prescription. Neither detector thermality nor dimensional analysis alone fixes the coefficient of the galactic law. Milgrom related the MOND acceleration scale to the de Sitter horizon through this temperature comparison, with a different interpolation. [2, 3, 4]

The capacity model supplies that coefficient through the tilt mode of the transverse frame. The baryonic field is its restoring stiffness and its inertia lever is $\ell = g\,c/H$, so its frequency

is $\omega = \sqrt{g_b/\ell}$. At the horizon temperature its gap is $\hbar\omega/k_B T_H = 2\pi\omega/H = \sqrt{g_b/a_0}$ with $a_0 = \epsilon c H$ and $\epsilon = g/(4\pi^2)$; for a lever in apparent-horizon radii the factor $4\pi^2$ is the square of the 2π in T_H. The construction connects the galaxy's transition scale to an external cosmological state without adjusting a coefficient for the high-redshift sample. The tilt-mode identification, its thermalization, and the lever are the matching inputs described in the theory; the lever coefficient is the ratio of the register enthalpy to a one-nat oriented record read at the Unruh temperature of the projected baryonic field, and that reading is the premise. [T1, section 15 and N.8]

The thermal variable is the occupation energy of the transverse response. The construction assigns no ordinary radiation temperature to an orbiting star.

3.3 The response law and its limits

The occupation energy of the tilt mode is

$$E = k_B T_H x, \qquad x = \sqrt{g_b/a_0}. \tag{18}$$

For a bosonic occupation number $n = 0, 1, 2, \ldots$, the single-mode partition sum is a geometric series:

$$Z_B = \sum_{n=0}^{\infty} e^{-nx} = \frac{1}{1 - e^{-x}}. \tag{19}$$

The mean occupation follows by differentiation,

$$\langle n \rangle = -\frac{d \ln Z_B}{dx} = \frac{1}{e^x - 1} = n_B(x). \tag{20}$$

Identifying the excess gravitational response with this occupation gives, in spherical symmetry,

$$g_{\rm obs} = g_b[1 + n_B(x)] = \frac{g_b}{1 - e^{-\sqrt{g_b/a_0}}}. \tag{21}$$

In the deep regime $n_B \simeq 1/x$ is the equipartition occupation of the mode, and at high acceleration its Wien suppression returns the Newtonian field. The Bose sum and spherical response are proved for the tilt-mode energy and readout. The carrier-action field equations also follow from compactly supported variations for a smooth interpolating function. For this response its derivative is singular at zero baryonic field, so the classical derivation applies on the regular nonzero-field branch; extension through field zeros requires a weak or regularized treatment. The square-root energy mapping and horizon contact remain physical matching conditions. [T4]

At $g_b \gg a_0$, the occupation is exponentially small and $g_{\rm obs} \simeq g_b$. At $g_b \ll a_0$, expansion of the denominator gives

$$g_{\rm obs} = \sqrt{a_0 g_b} + \frac{1}{2} g_b + O\left(\frac{g_b^{3/2}}{\sqrt{a_0}}\right). \tag{22}$$

For a pointlike baryonic source, $g_b = GM_b/r^2$, so the leading term implies

$$v_c^4 = GM_b a_0. \tag{23}$$

This is the baryonic Tully–Fisher scaling. In the proposed contact branch, its normalization evolves with $H(z)$. The isolated equilibrium response uses the cosmological contact at the epoch being modeled. A previously formed, subsequently stripped stock evolves through its own transport history.

A positive finite update can relax radial occupation while preserving each retained angular-bin mass. Lean verifies this separation, so radial adjustment need not erase angular memory. In

a spherical numerical example with a chosen 0.5 Gyr relaxation time, the force differs from the evolving target by at most 2.21% today and 12.89% earlier. Separate live tilted-stock realizations retain stellar tilt for eight Gyr at a fixed radial target. A common evolution with changing a_0, the required energy exchange, and converged nonspherical forces remains to be tested. [T1, O.3; G7]

3.4 Why the fitted intercept lies below the local value

The numerical curve in equation (13) has $a_0(0) = 1.230729\,10^{-10}\,\mathrm{m\,s^{-2}}$ and $a_0(1) = 2.203315\,10^{-10}\,\mathrm{m\,s^{-2}}$. The intercept of a line fitted over a positive-redshift interval is a different quantity from the value of the curved function at zero.

For points (z_i, f_i) with equal weights, minimizing $\sum_i [f_i - (A + Bz_i)]^2$ gives

$$B = \frac{\sum_i (z_i - \bar{z})(f_i - \bar{f})}{\sum_i (z_i - \bar{z})^2}, \qquad A = \bar{f} - B\bar{z}. \tag{24}$$

Using $z_i = 0.33, 0.34, \ldots, 1.44$ gives equation (14). The resulting intercept is 1.003 and the slope is 1.226; both are within the published marginal errors.

The curvature follows analytically. Write $u = 1 + z$ and $E = \sqrt{\Omega_m u^3 + \Omega_\Lambda}$. Then

$$\frac{d^2E}{dz^2} = \frac{3\Omega_m u\,[\Omega_m u^3 + 4\Omega_\Lambda]}{4E^3} > 0. \tag{25}$$

Across the stated interval, a single straight line averages the increasing slope and extrapolates below the true value at $z = 0$. The lower intercept therefore has a direct mathematical explanation within the proposed curve. Its precise size depends on the interval and weights.

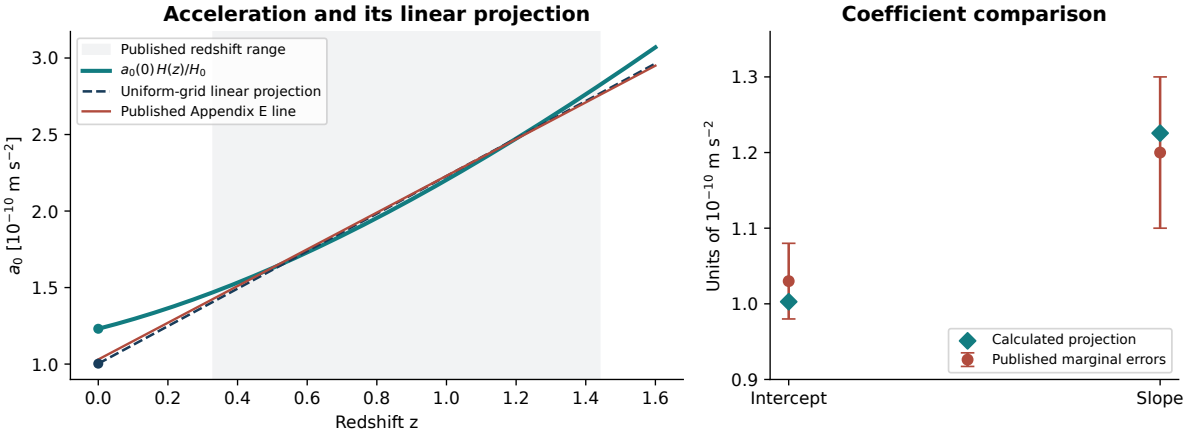


Figure 1: The calculated acceleration curve and its linear projection. The published line summarizes the Appendix E inference. The right panel shows marginal coefficient errors, with no assumption of independence. The uniform-grid projection is not a refit to the original galaxy data.

The individual coefficient differences are -0.54 and $+0.26$ times the quoted marginal errors. A joint significance requires the fitted intercept–slope covariance and the survey's effective redshift weighting. The two agreements are consequences of one curve, not independent probabilities to multiply.

3.5 A joint cosmological test

The prediction fixes the absolute function, including normalization and curvature. This comparison uses $\Omega_m = 0.315$ and $\Omega_\Lambda = 0.685$. A joint test must use the release model's fitted

$H(z)$ in equation (11), propagate its uncertainty, and fit the individual galaxies with their correlated baryonic and kinematic errors. Galaxy formation can also produce an evolving RAR in conventional halo models, so evolution alone does not distinguish the mechanisms.

3.6 Catalogue tests and the direction of a relaxation correction

The common cosmology was propagated to 73 KMOS3D, 21 KGES, and 169 KROSS objects at three radii. At the KMOS3D outer radius, the log-acceleration residual slope is -0.245 dex per unit redshift with constant a_0 and -0.313 with the common $H(z)$ law. The latter has median log-velocity residual -0.087 dex. The reference $z = 2.3$ example gives acceleration boosts about 2.02 versus 1.39 for the compared evolving and fixed scales. These are catalogue central-value diagnostics with specified baryon force models; they do not constitute raw-cube likelihoods. The 112-point reference projection remains distinct from a regression on the actual 79-object MUSE selection. The common cosmology has not removed the high-redshift residual trend. [U1, S01]

Passive relaxation cannot automatically supply a reduction. If $\dot{n} = \Gamma(q - n)$ with $\Gamma > 0$, $\dot{q} \leq 0$, and initially $n \geq q$, then n remains above the target. At fixed baryons a lag behind declining H therefore has the wrong sign to suppress an initially equilibrated response. Underoccupied assembly states require their own history. Relabeling an unchanged stock leaves its total stress unchanged; the tested adiabatic cosmological mass loss gives an inferred scale ratio 0.9183 from $z = 2.3$, compared with $H_0/H(2.3) = 0.2972$. These constraints identify the contact and preparation history as physical requirements. [D1]

For the selected two-pole quantum response, the noise enhancement $\mathcal{F}$ and mean delay $\bar{\tau}$ obey $\mathcal{F}H\bar{\tau} \geq 4$. The illustrative three-percent tracking condition at occupation argument $x = 1$ gives $\mathcal{F} \gtrsim 158$, requiring about 6.47×10^4 independent cells for three-percent force rms. Actual windows, correlations, and gravitational coupling remain unspecified, so this is a contact-model constraint. [D1; T1, N.8]

3.7 Stock selection reaches the response law

The response law is an equilibrium of the retained stock. The stock reaches it through a selection with three ingredients:

- **Occupation.** The response cell's occupation is the stock field in baryonic units, $n = g_R/g_b$.
- **Free energy.** Its excess free energy is the Bose relative entropy $D(n\|n_*)$. Its weight is fixed by equipartition with the excess field, $\mathcal{N}\Theta = g_b^2 n_*(1 + n_*)/4\pi G$.
- **Contacts.** Momentum-conserving stock contacts proceed when they lower the orbit-averaged free energy, which is the zero-temperature limit of detailed balance with the horizon bath.

A coupling that held the stock at the response density would pass the stock's kinematic mismatch d to the baryons as a force error $-dJ'(g_b)$, with $J' \simeq 5$ at 100 kpc. The stock therefore supports itself, and the selection acts slowly. [T1, O.3; G8, G9]

Spherical three-dimensional runs use abundance-matched NFW stock with an equilibrium Eddington distribution [11, 10], static spherical-equivalent baryons, and one to three contact proposals per orbit. They run for 12 Gyr. Offsets from the response law are rms over 5–200 kpc, and over 5–100 kpc for dwarfs:

Model	no contacts	8–12 Gyr	note
Milky Way, 4 runs	0.10	0.009–0.032	mismatch falls to the shot-noise level
Dwarf, 3 seeds	0.13	0.02–0.04	5–10% above the exact constrained minimum
Massive spiral, $6 \times 10^{12} M_\odot$	0.03	0.007–0.043	
Massive spiral, $2.5 \times 10^{12} M_\odot$	0.14	0.05	stock budget binds; still converging
Full Bose functional, MW		0.019–0.024	matches its Gaussian limit
Full Bose functional, dwarf		0.025–0.030	matches its Gaussian limit

Contacts conserve momentum and angular momentum to roundoff, and the energy ledger closes to $10^{-3} V_f^2$. The stock gains about $0.15 V_f^2$ per unit mass from the work register. The mismatch decreases overall, with fluctuations at the particle shot-noise level.

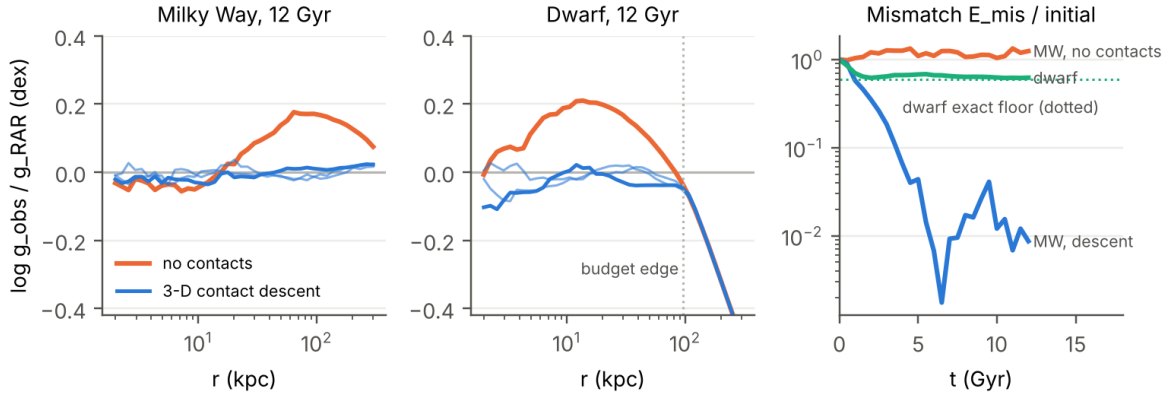


Figure 2: Offset from the response law after 12 Gyr without contacts (orange) and with contact selection (blue, three runs per model). Right: the mismatch functional relative to its initial value; the dotted line is the dwarf's exact constrained minimum.

The cohort ratio inside galaxies. Inside radius r the selected stock-to-baryon ratio equals the occupation $n_*(x)$, which rises outward. A Milky Way model has about 0.8 inside 10 kpc, 5.2 at 50 kpc, 11 at 100 kpc and 22 at 200 kpc, up to the budget edge, where the retained stock is exhausted. A galaxy that has lost most of its ancestral baryons from the counted region therefore shows a total of about 30 stock masses per present baryonic mass. The same stock, measured against the cohort's original baryons, holds the ratio k_C.

Weak lensing. An isolated finite stock bounds the excess surface density by $(1+S) g_{\rm bar}/\pi G$. A signal on the deep response law then requires $S \geq \frac{\pi}{4}\sqrt{a_0/g_{\rm bar}} - 1$: about 12 at 4.7×10^{-13}, 31 at 7.6×10^{-14} and 275 at $10^{-15}\,{\rm m\,s^{-2}}$. Stocks of about 30 fit the inner KiDS bins [29]. The carried stock ends with the support, where hot gas outweighs the condensed baryons, and the uncarried stock outside keeps the halo profile. A population forward model of the four isolated stellar-mass bins of [29], with 60 points, full covariance, the same hot gas and a common environment term, gives $\chi^2 = 140$ for the support-limited stock against 133 for truncated NFW halos when both fit a halo-mass shift and a concentration factor. Both leave 3–6σ residuals at 0.5–1.5 Mpc, where neighbouring structure contributes. [G8, G12]

4 The LMC and inherited gravitating stock

For $3 \times 10^9 M_\odot$ of LMC baryons, the retained-stock prescription gives

$$M(<30\,{\rm kpc}) = 5.0348 \times 10^{10} M_\odot, \qquad M(<50\,{\rm kpc}) = 8.2896 \times 10^{10} M_\odot. \tag{26}$$

Warren et al. infer $(4.7\text{–}5.1) \times 10^{10} M_\odot$ inside 30 kpc from the Orphan–Chenab stream. Brooks et al. infer $8.76^{+1.94}_{-1.77} \times 10^{10} M_\odot$ inside 50 kpc from outer-halo kinematics. The calculated masses agree with both quoted results. [5, 6]

In the same baryonic configuration, the tested QUMOND external-field calculations give only $(1.54\text{–}1.68) \times 10^{10} M_\odot$ of effective enclosed LMC mass inside 30 kpc. The difference comes from whether the system retains its previously established stock when it enters the Milky Way environment. [G1, G2]

4.1 From an acceleration law to an enclosed mass

Outside most of the LMC's baryons, use the spherical point-source approximation

$$g_b(r) = \frac{GM_b}{r^2}, \qquad r_M = \sqrt{\frac{GM_b}{a_0}}. \tag{27}$$

Equation (21) then gives

$$g_{\rm obs}(r) = \frac{GM_b}{r^2[1 - e^{-r_M/r}]}. \tag{28}$$

The effective enclosed mass is defined through $g_{\rm obs} = GM(<r)/r^2$, so

$$M(<r) = \frac{M_b}{1 - e^{-r_M/r}}, \qquad M_R(<r) = M(<r) - M_b. \tag{29}$$

The baryonic term and the extra stock are counted once each.

With $G = 4.30091 \times 10^{-6}\,\mathrm{kpc\,(km/s)^2} M_\odot^{-1}$, $a_0 = 3798.47\,\mathrm{(km/s)^2 kpc^{-1}}$, and the stated baryonic mass, $r_M = 1.84305\,\mathrm{kpc}$. At 30 kpc the exponent is $r_M/r = 0.061435$, and

$$M(<30) = \frac{3 \times 10^9}{1 - e^{-0.061435}} M_\odot = 5.0348 \times 10^{10} M_\odot. \tag{30}$$

At 50 kpc the exponent is 0.036861, giving the second mass in equation (26). A Plummer baryonic profile with scale 1.5 kpc changes these outer values only slightly.

In the deep regime,

$$M(<r) = \sqrt{\frac{M_b a_0}{G}}\, r + \frac{M_b}{2} + O(r_M/r). \tag{31}$$

The enclosed mass grows approximately linearly with radius, giving an approximately flat circular speed before the finite stock budget limits the profile. This also shows the main sensitivity: $M(<r) \propto M_b^{1/2} a_0^{1/2} r$. A fractional baryonic-mass error contributes about half as much fractional mass error at these radii. The central numbers do not imply negligible uncertainty in the baryons.

4.2 Why the budget refers to ancestral baryons

The initial commitment is bounded by $M_{b,L}/\epsilon$, where $M_{b,L}$ is the baryonic content of the Lagrangian source whose history is followed. In the linear-depletion benchmark used for the saved galaxy calculations, a full release removes the share η_*. Its carrier budget is

$$M_R \le \frac{1 - \eta_*}{\epsilon} M_{b,L}, \qquad C_R \equiv \frac{M_{R,\max}}{M_{b,\rm present}}. \tag{32}$$

Loss of gas or redistribution of baryons can make C_R much larger than $1/\epsilon$. A large present stock-to-star ratio is compatible with a finite initial commitment per ancestral baryon.

Writing $M_{b,L} = f_b M_h$ gives the manuscript's useful parameterization

$$C_R \simeq (1-\eta_*)(1-f_b)\frac{M_h}{M_{b,\mathrm{present}}}, \tag{33}$$

when $f_b/(1-f_b) \simeq \epsilon$. Here M_h identifies the ancestral assembly mass used for the budget. It is an astrophysical history input. Its use does not follow from the cell count alone.

The numerical LMC model uses $C_R = 33$, so the total mass cap is $(1+C_R)M_b = 1.02\times10^{11}M_\odot$. Equation (29) reaches that cap at

$$r_{\mathrm{cap}} = -\frac{r_M}{\ln[1-(1+C_R)^{-1}]} = 61.74\,\mathrm{kpc}. \tag{34}$$

The quoted 30 and 50 kpc predictions lie inside the cap. Their values are insensitive to its exact location. The composition-preserving allocation replaces $1-\eta_*$ by $e^{-\eta_*}$ for complete ensemble release, increasing this budget by 0.0455%. The reported masses and satellite trajectories retain their original benchmark convention. [T1, O.3; G1]

4.3 Persistence, transport, and tidal stripping

Retained stock has a positive density and an advected mass. A collisionless description uses a phase-space distribution $f_R(\mathbf{x},\mathbf{v},t)$ obeying

$$\frac{\partial f_R}{\partial t} + \mathbf{v}\cdot\nabla_x f_R - \nabla\Phi\cdot\nabla_v f_R = 0, \qquad \rho_R = \int f_R\, d^3v, \tag{35}$$

between formation, release, and carrier-reorganization events. Integrating over velocity gives a continuity equation for ρ_R. A change in the host's gravitational field can move the stock, distort it, or remove some of it from the satellite. It does not algebraically erase its mass.

A spatially uniform external acceleration mainly accelerates the carrier as a whole. Tides arise from the field gradient, represented locally by $\partial_i\partial_j\Phi_{\mathrm{host}}$. They can unbind the extended stock because its outer orbits are weakly bound. This is the physical difference between losing stock through transport and reducing a static effective gravity law because the external field changed.

The theory's equilibrium contact and its persistent-stock dynamics are different stages of the same proposed mechanism. The contact specifies a target profile for a coherent carrier; collisionless evolution follows that profile after infall. The manuscript also allows mass-conserving relaxation toward carrier-resolved equilibrium. The available satellite runs use retained stock with no subsequent contact relaxation. Its rate and its derivation from the native microscopic dynamics remain to be fixed.

4.4 The external-field comparison

In QUMOND, the field equation contains the nonlinear flux $\nu(|\mathbf{g}_N|/a_0)\mathbf{g}_N$, where the Newtonian field includes both host and satellite. The local calculation estimates the LMC's equivalent enclosed mass by subtracting the host-only flux:

$$M^Q_{\mathrm{LMC}}(<r) = -\frac{1}{4\pi G}\oint_{S_r}\Big[\nu\!\left(\frac{|\mathbf{g}_{N,L}+\mathbf{g}_{N,MW}|}{a_0}\right)(\mathbf{g}_{N,L}+\mathbf{g}_{N,MW}) - \nu\!\left(\frac{|\mathbf{g}_{N,MW}|}{a_0}\right)\mathbf{g}_{N,MW}\Big]\cdot d\mathbf{A}. \tag{36}$$

This is an effective gravitational mass, not a count of extra matter particles. The host subtraction matters because the flux is nonlinear.

The computation uses a $10^{10} M_\odot$ Hernquist bulge, a $5 \times 10^{10} M_\odot$ Miyamoto–Nagai disk, a $3 \times 10^9 M_\odot$ Plummer LMC, and a 50 kpc separation. Direct angular quadrature uses 96 Gauss–Legendre points and 192 azimuthal points. Re-execution gives

Response prescription	$M(< 30\,\mathrm{kpc})/[10^{10} M_\odot]$
Retained spherical stock	5.035
QUMOND, exponential RAR function	1.679
QUMOND, simple function	1.681
QUMOND, standard function	1.539

The tested external-field models fall a factor of roughly three below the cited 30 kpc inference. The difference persists across these three interpolation choices. A complete comparison would evolve the observed stream in each time-dependent model and compare its positions and velocities directly. The published mass posterior was inferred using gravitational models; it is not a raw model-free datum.

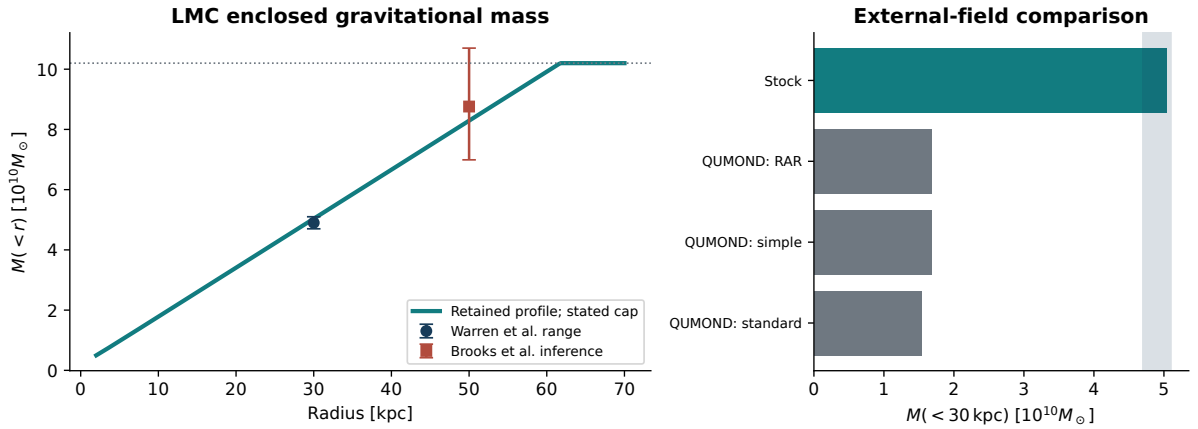


Figure 3: The retained LMC mass profile, its finite budget, and the recomputed QUMOND flux comparison. The 30 kpc bar represents the reported range of stream inferences, not a Gaussian one-sigma interval. The 50 kpc bar uses the published asymmetric uncertainties.

4.5 The reflex motion tests inertia

An extended gravitating component should also accelerate the Milky Way. Brooks et al. infer a reflex speed $38.6^{+8.3}_{-7.8}\,\mathrm{km\,s^{-1}}$ relative to halo stars at 100 kpc, jointly with the LMC and Milky Way masses. That observable tests the dynamical interaction, including the response of the outer halo. [6]

The available moving-LMC calculation produces a Milky Way acceleration corresponding to about $35\,\mathrm{km\,s^{-1}}$ over the last gigayear. This establishes the relevant scale. A direct comparison requires the same halo reference population, spatial selection, and velocity estimator as the observation. The present calculation also uses the usual asymmetric pair-force approximation, while the theory requires reciprocal forces from its joint source action. These distinctions prevent a precise reflex likelihood claim from being assigned to the saved run. [G4]

The physical implication of a successful reflex calculation would be strong: the extra gravity would participate in the momentum balance of the system. Agreement of an isolated rotation curve alone cannot establish that property.

4.6 Tidal dwarfs test which source owns the response

A tidal dwarf forms from material pulled out of a larger galaxy. Its gas and stars already belong to an ancestral source. The carrier rule assigns them to that inherited record while the parent remains dynamically coherent. Applying a new isolated equilibrium law to the dwarf's own baryons would count a new independent response at the wrong stage of its history.

Exact projectors act on disjoint active record labels. Smooth spatial weights used to distribute an effective source form a separate partition of unity; they carry no projector-idempotence requirement. If P is the active parent, its descendants contribute to the parent source before the nonlinear response is evaluated. Parent and child are not simultaneously counted as independent sources of an extra halo. A descendant in the parent's retained response also inherits its director, the orientation field that controls the transverse flux. [T1, section 15 and N.8]

The director's stiffness controls its response to a local tilt of the source. In the linearized description, a mode of wavenumber k has transfer factor

$$\frac{\delta \boldsymbol{n}(k)}{\delta \boldsymbol{n}_{\rm eq}(k)} = \frac{1}{1 + k^2 \xi^2}, \tag{37}$$

where ξ is the director coherence length fixed by the effective stiffness. A descendant samples the parent's coherent director. A separate carrier has its own stiffness and director. The inherited orientation suppresses the local transverse response that the independent-carrier calculation would allow. [G5]

The numerical proxy converts each resulting force field into a virial dispersion. For a stationary tracer distribution of mass M, define

$$W = -\int \rho(\boldsymbol{r})\, \boldsymbol{r} \cdot \boldsymbol{\nabla}\Phi \, d^3 r. \tag{38}$$

The scalar virial relation $2K + W = 0$ gives the direction-averaged one-dimensional dispersion $\sigma^2 = -W/(3M)$, and hence the boost $\sigma^2/\sigma_N^2 = W/W_N$. The force calculation, inherited orientation, and tracer profile determine this ratio. The observed gas rotation still requires a separate geometrical and dynamical interpretation.

The available NGC 5291 comparison uses three dynamical-to-baryonic mass ratios, 1.5 ± 0.7, 1.3 ± 0.5, and 1.2 ± 0.7, and a parent of $2 \times 10^{11} M_\odot$ at 60 kpc. The inherited response gives modest enhancements. Treating the dwarfs as separate isolated carriers gives much larger enhancements. [G1, G5][7]

Prescription	Mass enhancement	Reported χ^2 for three objects
Parent-carrier inheritance	1.5–2.0	0.5
Separate isolated carriers	3.6–4.9	48
Newtonian baryons	1.0	1.0

The calculation uses a virial-response proxy, with the enhancement read through σ^2/σ_N^2. It does not supply a resolved, non-equilibrium gas-dynamical fit to every dwarf. Its evidential force is the rejection of the separate-carrier prescription within this test. Newtonian baryons also fit these ratios, so these three objects do not uniquely select the capacity model or exclude conventional tidal dwarfs.

5 Cold dwarf remnants and tidal inheritance

Crater II combines a large projected half-light radius with a small stellar velocity dispersion. The moving-Milky-Way-plus-LMC calculation produces a surviving stock remnant with

$$\sigma_{\rm los} = 2.36\,{\rm km\,s^{-1}}, \quad R_h = 0.86\,{\rm kpc}, \quad M_* = 3.2\times10^5 M_\odot, \quad f_{\rm tail} = 0.23. \tag{39}$$

A second well-ranked state gives $(2.18\,{\rm km\,s^{-1}}, 1.24\,{\rm kpc}, 4.4\times10^5 M_\odot, 0.38)$. The comparison data used by the likelihood are $(2.7\pm0.3\,{\rm km\,s^{-1}}, 1.066\pm0.084\,{\rm kpc}, 3.2\pm1.0\times10^5 M_\odot, 0.23\pm0.08)$. The first state matches the dispersion, mass, and adopted tail statistic well, while its radius is about 19% low. The second is larger and has more stellar debris. [G3, G4]

Against the specified NFW halo ensemble, the moving-LMC comparison gives $B = 1.98\times10^8$ for 0.3-dex halo-mass scatter and $B = 2.62\times10^5$ for 0.5-dex scatter when the fractional model error is 10%. These values have been recomputed from the saved run outputs for this document. They quantify a preference for the stock construction within this numerical experiment.

5.1 Which satellites keep their stock

A satellite has its own cells only where its own baryonic field exceeds the host's. At the three-dimensional half-light radius the ratio of the Milky-Way law field to the dwarf's own law field is 5.5 for Sextans, 2.5 for Draco, 2.0 for Ursa Minor and 1.7 for Carina, which keep their pre-infall stock, and 0.98 for Sculptor, 0.38 for Fornax, 0.20 for Leo II and 0.09 for Leo I, which relax. Scored on the GravSphere central densities $\rho(150\,{\rm pc})$ [9], this assignment gives $\chi^2 = 10.4$ over the eight dwarfs, against 17.2 for retained stock throughout and 44 for relaxed stock throughout. Ten of the 256 retained or relaxed assignments reach $\chi^2 \le 10.4$; the best, 5.7, also retains Leo II. The classification holds for Milky-Way baryonic masses of $5\text{–}6\times10^{10}M_\odot$, cap ratios of 10–20, projected or three-dimensional radii, and Newtonian or law fields, with Sculptor on the boundary. [G12]

A dwarf whose main progenitor never crossed the release gate of Section 8 keeps its committed allocation, and its stock stays collisionless. Over 22 Local Group dwarfs with half-light masses from the Local Volume Database, relaxed masses from contact runs, and gate crossing from the mean accretion history of Fakhouri et al. (2010), this combined assignment gives $\chi^2 = 21.7$ at the reference gate of $17\,{\rm km\,s^{-1}}$ and 25.4 at $14\,{\rm km\,s^{-1}}$. ΛCDM halos with supernova heating at the published coreNFW calibration give 24.0, and NFW halos without heating give 30.0. Over plausible accretion histories, M31 baryonic masses and the two published Tucana dispersions, the combined assignment spans 18.5–29.2 and heating spans 23–25.5; these data do not fix the gate. [G12]

5.2 Why size and dispersion must be explained together

In a dispersion-supported system, random stellar velocities provide support against gravity. Under equilibrium and approximate spherical symmetry, a useful scale estimate is

$$M(<r_{1/2}) \simeq \frac{3r_{1/2}\sigma_{\rm los}^2}{G} \simeq \frac{4R_h\sigma_{\rm los}^2}{G}, \tag{40}$$

where $r_{1/2} \simeq 4R_h/3$ is the three-dimensional half-light radius. For the adopted Crater II values this gives roughly $7.2\times10^6 M_\odot$. The estimate is a guide to the required depth of the central potential; a tidally disturbed galaxy requires direct modeling. [8]

Reducing a halo's central mass can cool the stars, but its tidal response can also contract or destroy the stellar distribution. A successful model must retain a diffuse body while reducing its binding potential, and must leave the right amount of visible debris. A dispersion fit at an incorrect radius addresses only part of the problem.

5.3 Initial conditions and the physical stock profile

The stellar progenitor is a Plummer sphere,

$$\rho_*(r) = \frac{3M_*}{4\pi b^3}\left(1+\frac{r^2}{b^2}\right)^{-5/2}, \tag{41}$$

$$M_*(<r) = M_*\frac{r^3}{(r^2+b^2)^{3/2}}. \tag{42}$$

Its projected half-light radius is b. The progenitor's isolated stock is initialized from

$$M_R(<r) = \min\left\{M_*(<r)\left[\nu\left(\frac{GM_*(<r)}{a_0 r^2}\right)-1\right], C_R M_*\right\}, \tag{43}$$

with $\nu(y) = (1-e^{-\sqrt{y}})^{-1}$ and $C_R = 3000$. The differentiated density is tapered by $e^{-(r/12\,\mathrm{kpc})^2}$. The cap and taper are specified modeling inputs tied to the proposed history and its numerical realization. They are not independently measured halo properties of Crater II.

The central form explains why stock responds differently from a cusp. For $r \ll b$, the Plummer mass scales as r^3, so $g_b \propto r$. In the low-acceleration limit, $\nu - 1 \propto r^{-1/2}$. Consequently

$$M_R(<r) \propto r^{5/2}, \qquad \rho_R(r) \propto r^{-1/2}. \tag{44}$$

This is shallower than the r^{-1} inner density of an NFW halo. The outer stock is extended and weakly bound. Tides can remove substantial stock and lower the stellar velocity scale without forcing the same stellar contraction as in the tested NFW models.

Both stars and stock are initialized through isotropic Eddington distribution functions in the combined potential. This matters because the initial density profile alone does not specify stellar velocities. For relative potential $\Psi = -\Phi + \Phi(\infty)$ and relative energy $\mathcal{E} = \Psi - v^2/2$, the inversion can be written

$$f(\mathcal{E}) = \frac{1}{\sqrt{8}\,\pi^2}\frac{d}{d\mathcal{E}}\int_0^{\mathcal{E}}\frac{d\rho/d\Psi}{\sqrt{\mathcal{E}-\Psi}}\,d\Psi. \tag{45}$$

At a sampled radius, speeds are drawn with probability proportional to $v^2 f(\Psi - v^2/2)$. The code evaluates the inversion on a finite radial grid and clips negative numerical values of f to zero; equilibrium and resolution controls are needed to assess this approximation. Equilibrium phase-space sampling avoids using an arbitrary initial velocity dispersion as a hidden fit parameter. [G1, G3]

5.4 The orbit and the LMC perturbation

The calculation uses the present position, distance, radial velocity, and proper motion to infer a past orbit in each gravitational model. The adopted Crater II inputs are $D = 117.5\,\mathrm{kpc}$, $v_{\mathrm{los}} = 87.5\,\mathrm{km\,s^{-1}}$, and $(\mu_{\alpha*}, \mu_\delta) = (-0.07, -0.11)\,\mathrm{mas\,yr^{-1}}$. The stock Milky Way contains $6\times10^{10}M_\odot$ of baryons and a retained profile with $C_R = 14$.

The moving LMC carries $3\times10^9 M_\odot$ of Plummer baryons and its retained profile, capped at $1.02\times10^{11}M_\odot$ in total. The trajectory includes a specified Chandrasekhar friction term and the acceleration of the Milky Way. In this calculation Crater II's latest pericentre is 25.3 kpc, 1.68 Gyr before the present, in the stock host; the NFW comparison gives 24.9 kpc. The models use the same observed phase-space inputs, but their host potentials give distinct orbits. [G4]

Only first-pericentre stock progenitors survive in the sampled moving-LMC grid. Its best state begins with twice today's stellar mass and $0.7R_{h,\mathrm{obs}}$. The next state begins with four times today's mass and $R_{h,\mathrm{obs}}$. The agreement includes inference over progenitor mass, size, and infall age within a prior-weighted grid.

A rough tidal-radius estimate makes the mechanism clear. For a small satellite on a circular orbit of radius R in a spherical host,

$$r_t \simeq R \left[\frac{m(< r_t)}{[3 - d\ln M_H/d\ln R] M_H(< R)} \right]^{1/3}. \tag{46}$$

The real orbit is eccentric and the LMC makes the potential time dependent, so the simulation integrates the forces directly. Near pericentre, the strongest differential field removes the weakly bound outer stock. The retained stellar body then responds to a shallower potential. Its final dispersion is an output of this evolution.

5.5 What the numerical experiment actually resolves

The shared solver follows 40,000 stock or dark particles and 10,000 stellar particles with a one-Myr leapfrog step and 0.03 kpc softening. Self-gravity is computed through a monopole about the density centre. The external Milky Way and LMC fields are applied to the particles. This is a collisionless particle experiment with spherical internal force resolution, not a full unrestricted tree-gravity calculation.

The comparison uses NFW subhalos with a Moster stellar-to-halo mass prior and the Dutton–Macciò median concentration relation. Progenitor masses are one, two, or four times today's stellar mass; sizes are 0.7, 1.0, or 1.5 times the observed half-light radius, weighted by the stated size–mass prior. The infall-age grid counts pericentric passages. The enlarged halo grid includes masses far below the central abundance-matching expectation. [G3, G4][10, 11]

The NFW models approach the observed dispersion at inferred infall masses around $10^{7.4}$–$10^{7.7} M_\odot$, about two dex below the reference abundance-matching relation. Their radii then lie near 0.76–0.78 kpc and their dispersions near 3.2–3.5 $\mathrm{km\,s^{-1}}$. The two-dex statement describes the required prior displacement in this model family. It is not a universal proof that a galaxy could never have formed in such an object: gas retention, early cooling, concentration, and assembly history also matter.

5.6 Tail counts and the measurement procedure

The MAGIC photometric sample contains 162 candidate red-giant members, 37 of them in the tails, giving $37/162 = 0.2284$. The reported stream width is about 0.8°, and the tails extend beyond the observed footprint. These are population- and selection-dependent measurements. A candidate count is not automatically a complete fraction of all stellar mass. [12]

The grid likelihood uses $f_{\mathrm{tail}} = 0.23 \pm 0.08$ and a statistic counting stars beyond $3R_{h,\mathrm{obs}}$ within a three-dimensional radius of about 14.4 kpc. The central dispersion is measured from bound stars within $1.5R_h$ in projection. These definitions can be reproduced from the simulation, but the observing program does not have direct access to a bound-particle label or a spherical three-dimensional membership cut. A final comparison needs the photometric footprint, candidate selection, velocity errors, gradient subtraction, and foreground model applied to mock observations.

Independent spectroscopy now confirms cold central kinematics and provides a stream-to-body dispersion ratio. Limberg et al. report $2.51^{+0.33}_{-0.30}\,\mathrm{km\,s^{-1}}$ for the body and $5.74^{+0.98}_{-0.83}\,\mathrm{km\,s^{-1}}$ for the stream. Their own simulations find the dispersion ratio difficult for standard-concentration cusps, while cored or low-concentration halos can reproduce it. The stock model should predict this ratio and the stream width as additional observables. [13]

5.7 The strength and the scope of the model comparison

For an observable O_j, the grid uses an effective error $s_j^2 = \sigma_{j,\mathrm{obs}}^2 + (f_{\mathrm{mod}} O_{j,\mathrm{model}})^2$ for dispersion and radius. Its Gaussian likelihood includes the $1/s_j$ normalization. With fixed observational errors for stellar mass and tail fraction,

$$\mathcal{L}(\theta) \propto \prod_j \frac{1}{s_j} \exp\left[-\frac{(O_{j,\mathrm{model}}(\theta) - O_{j,\mathrm{obs}})^2}{2 s_j^2}\right]. \tag{47}$$

Destroyed states receive zero likelihood. The evidence is a discrete prior average, $Z = \sum_i w_i \mathcal{L}_i / \sum_i w_i$ for the normalized grid, with the documented Gaussian mass-prior quadrature for the extended halo tail. The Bayes factor is $B = Z_{\mathrm{stock}}/Z_{\mathrm{NFW}}$.

Fractional model error	With LMC, 0.3-dex mass scatter	With LMC, 0.5 dex
10%	1.98×10^8	2.62×10^5
20%	2.99×10^3	9.26×10^2
30%	1.55×10^2	9.42×10^1

The preference remains in the same direction throughout these sensitivity checks, while its numerical magnitude changes by several orders of magnitude. The quoted large factors require the 10% model-error assumption. The numerical errors, force approximation, orbit uncertainty, and observation mapping should determine that error budget independently.

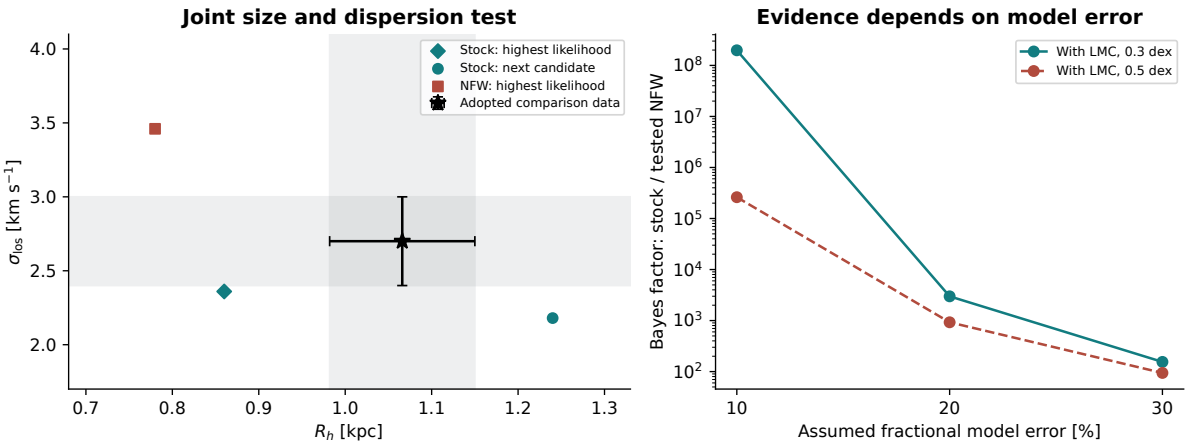


Figure 4: The moving-LMC Crater II comparison and its dependence on the assumed model error. The left panel emphasizes the joint size and dispersion constraint. The evidence calculation also uses stellar mass and the adopted tail fraction. The two halo priors use the same saved particle trajectories.

The archived record describes a hash-locked core protocol preceding the comparison campaign. The moving-LMC extension and enlarged mass priors are documented follow-ups. The present file hash matches the stored registration hash. A matching hash verifies file identity; by itself it does not independently certify the historical ordering of executions. [G3, G4, G6]

The tested alternative consists of cuspy halos at a prescribed concentration relation. It does not cover the full range of concentration scatter, feedback-induced cores, or every dark-matter model. Crater II therefore favors the stock model strongly over this specified alternative. It does not establish odds against the entire ΛCDM framework.

The moving-LMC satellite grid was reweighted with ten, twenty, and thirty percent model uncertainty. At thirty percent the Crater II stock/NFW evidence falls to 94–155 from 2.6×10^5–

2.0×10^8 at ten percent; Antlia 2 rises from 0.007–0.019 to 0.69–0.74. The grid does not implement the new conservative joint-stock contact or span broad cored-halo alternatives. Its evidence remains conditional on the specified histories, priors, and error budget. [U1, S14]

5.8 Antlia 2: a recently disturbed system still requires a successful mock observation

Antlia 2 has a measured dispersion $5.98^{+0.37}_{-0.36}\,\mathrm{km\,s^{-1}}$ after fitting a velocity gradient. The gradient is $5.72^{+0.60}_{-0.56}\,\mathrm{km\,s^{-1}}$ per degree. Ji et al. find evidence of tidal disturbance and show that its orbital interpretation requires the LMC-induced reflex of the Milky Way. [14]

In the available moving-LMC stock runs, five of the 27 states survive. The best bound remnant has $\sigma = 3.48\,\mathrm{km\,s^{-1}}$, $R_h = 2.41\,\mathrm{kpc}$, and $M_* = 9.4 \times 10^5 M_\odot$. Its most recent pericentre is about 0.79 Gyr ago. The NFW comparison can approach the adopted observables using low-mass halos; on the original bound-star likelihood it is favored over the stock by about 50–140. [G4]

The suggestion that Antlia 2 has not had enough time to reveal a settled internal dispersion can be made quantitative. Using its observed radius and dispersion,

$$t_{\mathrm{cross}} \sim \frac{R_h}{\sigma} \simeq 0.978\,\frac{2.92}{5.98}\,\mathrm{Gyr} \simeq 0.48\,\mathrm{Gyr}. \tag{48}$$

The elapsed time is only about 1.6 such crossing times; using the best stock remnant gives a crossing time near 0.68 Gyr. A dynamically disturbed state is plausible. This concerns time since disturbance, not the age of its stellar population.

The available approximate mock analyses show that unbound members and velocity-mixture choices can move the inferred dispersion substantially. They even change the relative ranking of the tested models. That is a reason to use the observed selection and membership procedure. A quantitative explanation of the measured 6 $\mathrm{km\,s^{-1}}$ remains to be established. The collisionless integrations already contain the tidal evolution; a successful non-equilibrium account must show how that evolution produces the measured dispersion, gradient, density profile, and membership sample together. Antlia 2 remains an unresolved test of the current construction. [G6]

5.9 What the Crater II result changes

The successful stock trajectories show that a diffuse cold remnant can emerge from a specified positive mass component tied to baryonic history. This challenges the assumption that the excess gravity of such a dwarf must be recalculated from its present stars, or must be supplied by a standard-concentration NFW subhalo with the adopted abundance prior. The mechanism is explicit: outer stock is stripped, the central potential weakens, and a substantial stellar body survives.

The next discriminants are the full stream structure, the velocity distribution including unbound stars, and an independently controlled model-error budget. They test the same history that produced the central dispersion, so they can constrain the model without introducing a new halo normalization for each observable.

6 Common coefficients and additional discriminants

6.1 A common CMB-to-galaxy normalization

The bare commitment ratio and galactic scale obey

$$\left(\frac{\rho_{C,i}}{\rho_{b,i}}\right)_{\mathrm{bare}} = \frac{1}{\epsilon}, \qquad a_0(z) = \epsilon c H(z), \qquad a_0(z)\left(\frac{\rho_{C,i}}{\rho_{b,i}}\right)_{\mathrm{bare}} = cH(z). \tag{49}$$

The fitted formation convention includes the mass-conversion factor $f_m = 5.333\epsilon$, and release leaves the homogeneous survival factor $s(z) = e^{-\eta_* F(z)}$. On the undepleted background branch, the observable stock ratio therefore satisfies

$$a_0(z)\frac{\bar{\rho}_C(z)}{\bar{\rho}_b(z)} = f_m s(z) c H(z). \tag{50}$$

The standard linear-depletion configuration gives $H_0 = 68.458\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $a_0(0) = 1.25005 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, about 1.6% above the reference value. The exponential-allocation fit with massless neutrinos, linear lensing and a low-redshift supernova offset gives $H_0 = 68.93885586416\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $a_0(0) = 1.25883 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, about 2.3% above the reference value. Each value follows from its fitted H_0 without a separate fit of a_0. This is a conditional relation between sectors. Cosmological abundance and H_0 are inferred together; galaxy mass calibration limits the elementary empirical comparison to roughly twenty percent. The tilt-mode lever and mass convention retain their physical matching assumptions. [U1, S02; T4–T5]

6.2 Newton's constant and charged-lepton masses

The selected particle-coefficient route gives $G = 6.6742890813 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$, -0.073 quoted CODATA standard deviations; the muon ratio is 206.768280237, at -0.535 quoted standard deviations. Its tau mass is 1776.9188 MeV against 1776.93 ± 0.09 MeV. Of the 401 tested routing integers, only the integer 7 lies within two quoted standard deviations for the stated route; of 43 tested Farey candidates, only 2/7 lies within one. These finite-menu uniqueness checks do not quantify the historical search over possible formulas. The measured Newton scale and marked-formula residuals were known during development. [U1, S11][43]

6.3 Halo tilt and a common Milky Way–LMC history

Three eight-Gyr halo phase realizations retain late stock tilts of 39.58°–42.36° and stellar tilts of 21.38°–22.33°. Rounding and precession persist. These are repeatable outcomes of the specified kinetic model, with force convergence and stationary formation still unresolved. The aligned-branch failure remains part of the comparison. A common Milky Way–LMC–stream fit remains open. The archived Galactic masses inside 50 kpc exceed the cited constraint; Section 10.3 quantifies that mismatch. [U1, S13; G3–G7]

6.4 Old-disk vertical gravity

At the exponential fit's rounded $a_0 = 1.2588 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, the reduced old-disk solver gives NGC 628 $\chi^2 = 2.04$ for five dispersions and a -0.21σ radial–vertical mass offset. The approximate algebraic QUMOND comparison gives $\chi^2 = 8.95$ and -2.91σ. The small normalization change preserves the earlier pattern. Gas-profile, inclination, and scale-height uncertainties remain, and a round conventional halo can give similar vertical gravity. [U1, S15]

6.5 Wide binaries and Solar-System null tests

With the parent-carrier rule, a wide binary has no independent capacity halo. A radial Galactic-tide estimate for one to two solar masses at 10–30 kAU gives $\Delta v/v \simeq 1.1 \times 10^{-5}$–$6 \times 10^{-4}$. The range depends on separation, mass, and orientation; a universal 10^{-4} ceiling is unwarranted. The selected Solar quadrupole $Q_2 = 0$ is 0.89 quoted standard deviations from the Cassini central value. No full Gaia selection-and-contamination likelihood has been performed. [U1, S10]

6.6 Stock selection discriminants

The selection makes three tests that a fixed force law and a halo population without selection do not share:

- for a fixed baryonic target and a_0, relaxation reduces residuals from the response law with time since the last major accretion; longer outer orbits can retain larger residuals at a given age. Evolution of the target adds a history-dependent lag;
- isolated ellipticals follow the law out to the radius where their hot gas outweighs their stars;
- the ratio $a_0(z)/a_0(0) = H(z)/H_0$ applies to relaxed systems, and a 3σ distinction from a conventional rise of about three at $z = 2.3$ [33] needs a total uncertainty near 4.5%.

The dwarf runs reach the law at the mass of rotationally supported dwarfs. The GravSphere sample at $10^4 < M_{\rm bar}/M_\odot < 10^{7.5}$ lies above the extrapolated relation [32], and the selection does not yet extend to that regime. [G8, G9]

Part II. Recovery of established phenomenology

7 Established tests from galaxies to gravitational waves

The local acceleration relation, lensing spectra, and tensor propagation test whether the effective branches recover measured phenomena already described by MOND or conventional gravitating matter and general relativity. Their selections and covariance treatments remain separate. Overlapping catalogues and spectra require a joint likelihood before their evidence can be combined. [U1]

7.1 Local RAR and baryonic Tully–Fisher

For 130 selected SPARC galaxies with 2710 radial points, profiling the stated stellar mass-to-light, distance, and inclination priors gives a common-a_0 penalized score of 11258.0. Adding a $5\,\mathrm{km\,s^{-1}}$ velocity floor gives 3433.2, with median per-galaxy χ^2 per point of 1.54. The sensitivity to the adopted velocity errors is substantial. This is a diagonal-error algebraic-RAR check; it does not replace the full director-field calculation or account for all correlated noncircular motions. For 100 flat-velocity galaxies, the baryonic Tully–Fisher log residual has median 0.020 dex and rms 0.067 dex. [U1, S03][39]

7.2 Galaxy weak lensing

The isolated-lens audit distinguishes the older KiDS release with its covariance from the updated published table. At the reference a_0, the latter gives a diagonal statistic of 10.06 for fifteen bins, or 5.99 for seven inner bins, including the tabulated statistical and interpolation/extrapolation uncertainties. Its full table covariance was unavailable, and its published $H_0 = 73$ convention was retained. The two reductions overlap and provide no independent combined evidence. [U1, S04][40]

7.3 Primary CMB spectra and projected Weyl power

The positive-allocation cosmology of Section 8 was also evaluated against 135 ACT primary-spectrum bins, profiling the stated calibration at fixed cosmological parameters. The score is 162.73 against 162.02 for the reference ΛCDM fit. The ten-bin ACT lensing comparison gives 16.21 against 17.86. A higher-accuracy lensing calculation changes the bandpowers by 0.013%. These are separate checks; adding their scores to Planck would require the appropriate overlap cuts and joint covariance. [U1, S05–S06][41]

The ACT lensing calculation tests projected Weyl power. A DESI full-shape clustering likelihood with survey windows, bias, and redshift-space distortions, and a nonlinear weak-lensing likelihood, remain open. [U1, S06]

7.4 Gravitational slip and the lensing–velocity statistic

The lensing–velocity statistic E_G depends on the Weyl potential and the baryon velocity response. The calculation gives 0.4375 at $z = 0.316$ and 0.3862 at $z = 0.555$, nearly constant over $k = 0.005$–$0.15\,\mathrm{Mpc}^{-1}$. The cited ACT–BOSS estimates are $0.49^{+0.14}_{-0.11}$ and $0.31^{+0.06}_{-0.05}$. The no-slip identity alone does not determine this observable. In the conserved-fluid implementation, CAMB's ordinary `delta_tot` omits part of the bookkeeping capacity component; a physical total-matter σ_8 must be reconstructed independently of the arbitrary reference-dust split. [U1, S09][42]

7.5 Tensor polarization and propagation

For a constant Einstein coefficient and minimally coupled scalar matter, the tensor quadratic action has two positive-energy luminal modes. Public GWTC-4 polarization comparisons give combined $\log_{10}\mathcal{B}_{S/T} = -14.72$ and $\log_{10}\mathcal{B}_{V/T} = -5.33$ for pure scalar and vector alternatives. Mixed tensor–scalar and tensor–vector alternatives have values -0.20 and $+0.10$, respectively. This supports compatibility with the minimal tensor branch and leaves mixed sectors weakly constrained; it supplies no distinctive preference over GR. The native tensor continuum and a map from higher-curvature operators to propagation or Lorentz-violation bounds remain open. [U1, S12; D1][44]

8 Dark energy, distances, and the primary CMB fit

The geodesic-clock release model gives fits close to ΛCDM on Planck primary CMB spectra, DESI DR2 baryon acoustic oscillations, and DES-Dovekie supernova distances. The standard linear-depletion configuration uses $\sum m_\nu = 0.06\,\mathrm{eV}$, the Dovekie likelihood as released, nonlinear lensing from each model's perturbations, and a free Planck calibration with prior 1 ± 0.0025. It gives $\Delta\chi^2 = +0.85$, with five cosmological parameters for release and six for ΛCDM. The configuration with massless neutrinos, fixed calibration, nonlinear lensing and a free supernova offset below $z = 0.1$ gives $\Delta\chi^2 = -0.39$. [C6]

The exponential positive-allocation fit uses massless neutrinos, fixed calibration, linear lensing and the low-redshift offset. It gives $\Delta\chi^2 = -0.5152$; a separate archived linear-depletion fit with the same likelihood settings gives -0.5186. Their relative scores differ by about 0.0034, with different numerical accuracy settings contributing to the comparison. Their ΛCDM reference is 2228.316, against 2228.359 for the one-harness comparison of Section 8.6. Its conversion fraction is $\eta_* = 0.0298668$, its initial commitment abundance is tied to baryons, and one parameter sets the mixing time. [C4, C5]

The hydrodynamic calculation tests the release normalization. With the retention gate taken from the halo occupation of feedback-resolved simulations and no input adjusted to the vacuum density, it gives $\Omega_\Lambda = 0.709 \pm 0.02$ against the likelihood value 0.707. With the reionization redshift taken from the fitted optical depth, that frozen history fits the CMB, BAO and supernova data with four cosmological parameters at $\Delta\chi^2 = +1.07$ against six-parameter ΛCDM(Section 8.9).

8.1 What the expansion measurements constrain

In a homogeneous universe, the scale factor $a(t)$ measures relative expansion and $H = \dot{a}/a$. Nonrelativistic matter of conserved comoving mass has density proportional to a^{-3}. A vacuum component with pressure $p_\Lambda = -\rho_\Lambda$ has constant density when it exchanges no energy with other components. Throughout this section, energy-density equations use $c = 1$.

The observations constrain this expansion through complementary physical scales. Supernovae supply relative luminosity distances,

$$D_L(z) = (1+z)c\int_0^z \frac{dz'}{H(z')} \tag{51}$$

in a spatially flat universe. Their absolute magnitude and calibration must be inferred along with the cosmology. Baryon acoustic oscillations compare the transverse distance D_M and radial distance $D_H = c/H$ with the sound horizon r_d. The CMB fixes the early baryon and gravitating matter densities, the primordial fluctuation spectrum, and the angular size of the acoustic scale. A competitive joint fit must reproduce the early acoustic physics as well as the late distance relation. [15, 22]

In the capacity model, committed material exists before the acoustic epochs and acts approximately as collisionless matter. Subsequent structure formation transfers part of its energy into the vacuum sector. This timing allows an early gravitating component and a late accelerating component to share an origin. The theory specifies an effective formation map and pre-acoustic epoch in O.15. Its microscopic formation vertex remains open; the abundance and subsequent release retain the stated ledger assumptions.

8.2 The record that mixing can destroy

An assembled baryonic object has a collective source identity: its constituents can be associated with a particular progenitor. Let A denote that progenitor label and Y_t a physically specified resolution of the baryonic state at time t. The mutual information

$$I(A;Y_t) = \sum_{A,Y} p(A,Y;t) \ln \frac{p(A,Y;t)}{p(A)p(Y;t)} \tag{52}$$

measures how much the resolved state tells us about the progenitor. For a fixed cohort and label alphabet, the normalized identity loss is

$$K_{\mathrm{mix}}(t) = 1 - \frac{I(A;Y_t)}{H(A)} = \frac{H(A \mid Y_t)}{H(A)}. \tag{53}$$

It is zero when the progenitor can be recovered perfectly and approaches one when the resolved state carries no information about that label.

Irreversible mixing gives this definition a dynamical basis. Suppose the mass fraction $c_A(\boldsymbol{x},t)$ of each label is advected by the same velocity field and diffuses with the same positive semidefinite tensor D. For a closed cohort of gas mass M_g, fixed label fractions, and vanishing boundary flux, the resolved continuum information obeys

$$\frac{dI}{dt} = -\frac{1}{M_g} \sum_A \int \frac{\rho}{c_A} (\boldsymbol{\nabla} c_A)^T D (\boldsymbol{\nabla} c_A)\, d^3x \le 0. \tag{54}$$

For the resolved continuum,

$$I = \frac{1}{M_g} \sum_A \int \rho c_A \ln(c_A/\bar{c}_A)\, d^3x. \tag{55}$$

Inserting the advection–diffusion equation into its time derivative and integrating the diffusion term by parts gives Eq. (54). Advection transports labels; diffusion reduces the gradients that distinguish their origins.

The theory identifies this loss of a collective baryonic source label with the release of its associated commitment. Histories exported to other registers can persist: the relevant loss is the supporting label in the baryonic carrier. A permutation of identical streams supplies no distinguishable source record by itself. The release trigger therefore depends on resolved source identity and an irreversible physical process. Equation (54) establishes monotonicity under its stated transport assumptions. Measurements in changing spatial bins also contain coarse-graining effects, which must be separated from irreversible mixing. [T1, O.7–O.14]

8.3 From retained wells to a release history

The assembly calculation follows the fraction $F_Z(t)$ of matter that has first entered a well able to retain its baryons. At virial radius R, define $V^2 = GM/R$ and $M = (4\pi/3)\Delta_{\mathrm{vir}}\rho_{\mathrm{crit}}R^3$, with $\rho_{\mathrm{crit}} = 3H^2/(8\pi G)$. Eliminating R gives

$$M_{\mathrm{crit}}(z) = \frac{V_{\mathrm{crit}}^3}{GH(z)\sqrt{\Delta_{\mathrm{vir}}(z)/2}}, \qquad t_{\mathrm{dyn}}(z) = \frac{R}{V} = \frac{1}{H(z)\sqrt{\Delta_{\mathrm{vir}}(z)/2}}. \tag{56}$$

The current saved assembly calculation uses $V_{\rm crit} = 17\,{\rm km\,s^{-1}}$ before $z_{\rm re} = 7.66$ and $30\,{\rm km\,s^{-1}}$ afterward. These thresholds represent gas retention before and after photoheating. They are astrophysical inputs to the present construction.

The excursion-set approximation represents the smoothed initial density contrast by a stochastic trajectory $\delta(S)$, where $S(M) = \sigma^2(M)$ increases toward smaller masses. Collapse requires crossing $B(z) = 1.686/D(z)$, with growth factor $D(0) = 1$. If $\mathcal{M}(S) = \max_{s\le S}\delta(s)$, the absorbed first-entry fraction is

$$F_Z(z) = \Pr\left[\max_{z'\ge z}\left\{\mathcal{M}[S(M_{\rm crit}(z'))] - B(z')\right\} \ge 0\right]. \tag{57}$$

The maximum over earlier times retains a successful entry even if the retention threshold subsequently rises. Each trajectory contributes once. The saved implementation uses 3×10^5 trajectories, Brownian-bridge crossing corrections, and an Eisenstein–Hu variance normalized to $\sigma_8 = 0.810$. It gives $F_Z = 0.0694$ at $z = 6$, 0.2411 at $z = 3$, and 0.6700 today. [C2]

A cohort entering at t_f then loses its label over a finite time. The fitted kernel is

$$K_{\rm mix}(t - t_f; t_f) = 1 - \exp\left[-\frac{t - t_f}{f\, t_{\rm dyn}(t_f)}\right], \qquad t \ge t_f. \tag{58}$$

Summing over cohorts gives

$$F_{\rm rel}(t) = \int^t K_{\rm mix}(t - t_f; t_f)\, dF_Z(t_f). \tag{59}$$

The dimensionless f is the sole release-history parameter in the cosmological fit. The exponential approximates unresolved transport. The closure calculation fixes η_* under the common energy-conversion matching. The mixing time, retention thresholds, and physical label resolution require separate inputs.

8.4 Energy transfer and the vacuum density

Let $\rho_{c0}^{\rm wb}$ be the present density the committed component would have under dust dilution without release. The linear-depletion benchmark used in the likelihood has

$$\rho_C(a) = \rho_{c0}^{\rm wb} a^{-3}\left[1 - \eta_* F_{\rm rel}(a)\right]. \tag{60}$$

The transfer rate per physical volume is $Q = \eta_*\rho_{c0}^{\rm wb} a^{-3}\dot F_{\rm rel}$. Conservation gives

$$\dot\rho_C + 3H\rho_C = -Q, \tag{61}$$

$$\dot\rho_\Lambda = Q, \qquad p_\Lambda = -\rho_\Lambda. \tag{62}$$

With zero initial vacuum term, integration yields

$$\rho_\Lambda(a) = \eta_*\rho_{c0}^{\rm wb}\int_{a_i}^{a} \widetilde a^{-3}\, dF_{\rm rel}(\widetilde a). \tag{63}$$

The combined stress-energy is conserved. The a^{-3} weighting gives early release a larger contribution to the present vacuum density. The history is an ensemble expectation in the statistically homogeneous state, evaluated on the local source clock. A realized local mixing event does not instantaneously update that prescribed expectation.

Composition-preserving allocation instead gives

$$\rho_C^+(a) = \rho_{c0}^{\rm wb} a^{-3} e^{-\eta_* F_{\rm rel}(a)}, \tag{64}$$

$$\rho_\Lambda^+(a) = \eta_*\rho_{c0}^{\rm wb}\int_{a_i}^{a} \widetilde a^{-3} e^{-\eta_* F_{\rm rel}(\widetilde a)}\, dF_{\rm rel}(\widetilde a). \tag{65}$$

Its internal components stay nonnegative on the domain of positive total stock. A prescribed homogeneous drain can still exhaust an underdense region; this total-stock boundary requires a physical completion. At the reference history, the homogeneous comoving loss is 1.9618%, compared with 1.9813% in the linear benchmark. Section 8.6 gives a matched fit of the weighted vacuum integral and its geodesic perturbations with massless neutrinos, linear lensing and a low-redshift supernova offset. [T1, O.3; T4]

The total capacity budget ties the would-be committed abundance to the baryons. The bare coefficient is $1/\epsilon = 5.32068$; the formation prescription's free-nucleon mass convention, including its helium correction, gives the adopted $\rho_{c0}^{\mathrm{wb}}/\rho_{b0} = 5.333$. This tie assumes that the relevant charge and commitment were established before the CMB acoustic epochs. Release slightly reduces the actual present dust-to-baryon ratio. [T1, O.3 and O.13; C3]

Write $\omega_i = \Omega_i h^2$, with $H_0 = 100h$ in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. For the linear-depletion benchmark, spatial flatness gives

$$h^2 = \omega_r + \omega_b + \omega_c^{\mathrm{wb}}[1 - \eta_* F_{\mathrm{rel}}(1)] + \eta_* \omega_c^{\mathrm{wb}} I_{\mathrm{rel}}, \qquad I_{\mathrm{rel}} = \int_{a_i}^{1} a^{-3}\, dF_{\mathrm{rel}}. \tag{66}$$

Here h is calculated from the release history and densities. The six-parameter reference model fits $\omega_b, \omega_c, h, A_s, n_s, \tau$. The tied release model fits $\omega_b, f, A_s, n_s, \tau$. Shared supernova calibration nuisance parameters are handled identically in both models.

8.5 A covariant effective description

The current theory supplies a constrained clock field ϕ whose gradient follows the committed dust. With signature $(-,+,+,+)$, define $X = -\frac{1}{2}\nabla_\mu\phi\nabla^\mu\phi$ and

$$S_{C\Lambda} = \int d^4x\sqrt{-g}\left[\lambda\left(X - \tfrac{1}{2}\right) - V_F(\phi)\right]. \tag{67}$$

Variation of λ imposes a unit timelike clock gradient. Writing $u_\mu = -\nabla_\mu\phi$ gives

$$T_{\mu\nu}^{C\Lambda} = \lambda u_\mu u_\nu - V_F g_{\mu\nu}, \qquad \rho_C = \lambda, \quad \rho_\Lambda = V_F. \tag{68}$$

The prescribed potential is the integrated release history in Eq. (63), expressed as a function of the background clock and fixed before variation. The clock equation produces the energy exchange between dust and vacuum. It also specifies perturbations: in Newtonian gauge the constraint gives $\delta\dot{\phi} = \Psi$, and $\delta\rho_\Lambda = \dot{V}_F\delta\phi$.

The action fixes a local effective evolution for the prescribed history. In the geodesic clock's comoving synchronous gauge, $\delta\rho_\Lambda = 0$ and

$$\dot{\delta}_C = -\tfrac{1}{2}\dot{h}_s + \frac{\dot{\rho}_\Lambda}{\rho_C}\delta_C, \tag{69}$$

where h_s is the synchronous metric trace. An exact linear fluid representation subtracts conserved reference dust $\rho_{\mathrm{ref}} \propto a^{-3}$ on the same flow and defines

$$\rho_D = \rho_C - \rho_{\mathrm{ref}} + \rho_\Lambda, \qquad p_D = -\rho_\Lambda, \qquad c_{s,D}^2 = 0. \tag{70}$$

The conserved components reproduce the physical clock stress and metric perturbations. Choosing $\rho_{\mathrm{ref}} < \rho_C$ keeps the residual enthalpy positive. Numerical changes of this arbitrary split leave lensing power invariant to 1.16×10^{-7} in the tested range; an independent subhorizon growth calculation agrees within 9.2×10^{-7} in the tested power ratio. These are saved numerical checks. The full cosmological perturbation derivation is outside the current Lean proofs. [C4; T4]

8.6 The matched linear-lensing likelihood

The saved exponential-allocation fit has

$$\omega_b = 0.02237932, \qquad f = 0.27424478, \qquad h = 0.68938856, \tag{71}$$

$$\ln(10^{10} A_s) = 3.04585067, \qquad n_s = 0.96662931, \qquad \tau = 0.05661582. \tag{72}$$

Both local optimizations converged. Higher-accuracy evaluations of their endpoints give

Model	CMB	BAO	Supernovae	Total χ^2
Geodesic release	589.078	11.707	1627.016	2227.801
ΛCDM	590.215	11.191	1626.909	2228.316

The release and reference fits have $H_0 = 68.9389$ and $68.6558\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, respectively. Both use massless neutrinos, fixed calibration, linear lensing and the supernova likelihood with the low-redshift offset described below, and their difference is $\Delta\chi^2 = -0.515$. [C4, C5]

One harness evaluates both models on the same likelihood, with the linear-depletion direct geodesic release and its matched lensing. It reproduces the ΛCDM values 2226.767 (nonlinear lensing) and 2228.359 (linear lensing) of the pipelines in this section, and its Gauss–Newton minima are converged to 0.005 in χ^2:

Supernovae, neutrinos, lensing	ΛCDM	Release	$\Delta\chi^2$
low-z offset, $\sum m_\nu = 0$, linear	2228.359	2227.777	−0.58
low-z offset, $\sum m_\nu = 0$, nonlinear	2226.767	2226.380	−0.39
Dovekie, $\sum m_\nu = 0.06\,\mathrm{eV}$, nonlinear	2239.031	2239.880	+0.85

The first two rows fix the Planck calibration at unity. The last row frees it with the prior 1 ± 0.0025; its release fit has $H_0 = 68.458\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. Its difference splits into −1.54 from the CMB, +1.98 from BAO, +0.83 from supernovae and −0.42 from the priors. With $\omega_c = 5.333\,\omega_b$ the release model cannot offset the neutrino mass by lowering ω_c, which sets the BAO cost. Counting five versus six cosmological parameters gives $\Delta\mathrm{AIC} = -1.15$ on that row. This conditional parameter-count comparison supplies neither a posterior odds ratio nor a penalty for the choice of release prescription. [C6]

The archived PPF calculation with a fixed lensing rescaling gave 2227.329 against its matched 2226.77 reference. Within that pipeline, fixing $f = 0.5$ or 0.75 raised χ^2 by 45.87 or 189.62 relative to its reference, and the reported width was $f \simeq 0.28 \pm 0.03$. Those sensitivity results belong to that pipeline; the absolute statistics cannot be compared across the two lensing treatments. Its weighted-integral target $I_{\mathrm{rel}} \simeq 94.2$ remains the reference for the archived hydrodynamic comparison below. [C1, C2]

The saved exponential fit and the massless-neutrino comparisons use Planck 2018 `plik_lite` high-ℓ TT, TE, and EE, supplemented by two low-ℓ temperature bins and a Gaussian optical-depth prior $\tau = 0.0506 \pm 0.0086$. Calibration $y_{\mathrm{cal}} = 1$ is fixed. The BAO piece uses thirteen distance measurements from seven DESI DR2 tracer/redshift samples, retaining the tabulated within-sample correlations. The sound horizon and distances are computed by CAMB. [C1, C2, C4]

The supernova archive used by the low-redshift-offset fits contains the DES-Dovekie recalibration data and covariance. The code profiles a common magnitude offset and an additional offset for $z < 0.1$. Explicitly, if r is the residual vector, W the inverse covariance, and $X = (\mathbf{1}, \mathbf{1}_{z<0.1})$, the minimized statistic is

$$\chi^2_{\mathrm{SN}} = r^T W r - (X^T W r)^T (X^T W X)^{-1} (X^T W r). \tag{73}$$

Both cosmologies receive the same freedom. The present best-fit record does not save the fitted low-redshift offset, so a numerical offset is not assigned to it here. Supernova calibration remains part of the joint inference. [23, 24][C1]

8.7 Accumulation at high redshift and dominance at late time

The fitted exponential-allocation history gives

$$F_{\mathrm{rel}}(0) = 0.6635663, \qquad z_{1/2} = 7.9278, \tag{74}$$

$$\frac{\rho_\Lambda(2)}{\rho_\Lambda(0)} = 0.9682403, \qquad e^{-\eta_* F_{\mathrm{rel}}(0)} = 0.9803765. \tag{75}$$

Here $z_{1/2}$ is the epoch at which the vacuum density first reaches half its present value. About 65.8% is present by $z = 6$. The remaining stock has lost 1.96235% of its initial comoving mass, and the drain between $z = 1$ and today is only 0.49347% of that initial stock. The vacuum integral weights release by a^{-3}, so early release contributes disproportionately to the accumulated density. These values condition on the fitted kernel and reference assembly table used in the history figure below. [C5]

Early accumulation is compatible with late dynamical importance. Matter density was much larger at high redshift and continues to dilute as the universe expands. A vacuum density already close to its final value can therefore remain a small fraction of the total for a long interval. The coincidence problem becomes a question about the normalization and timing of assembly, retention, and release. Equation (63) connects these quantities quantitatively.

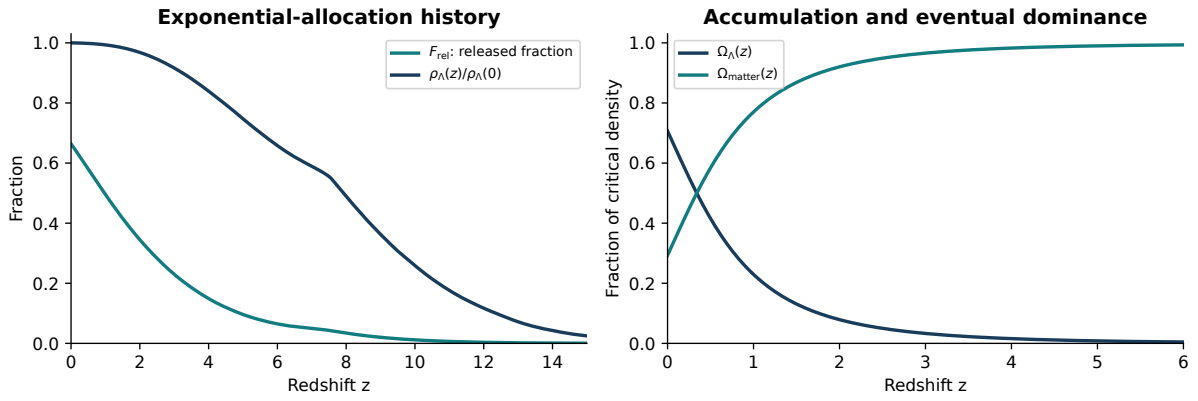


Figure 5: The fitted exponential-allocation release history from the table. The released fraction counts cohorts; the vacuum density weights their release by the physical matter density at that epoch. The right panel shows why a substantial early vacuum density becomes dynamically important only much later. These curves condition on the fitted kernel and fixed reference assembly history.

8.8 What the hydrodynamic measurement finds

The archived Gadget-4 calculation measures progenitor information during self-gravitating assembly. It follows 2×96^3 particles in a $0.75\,h^{-1}$ comoving Mpc box, from initial conditions at $z = 63$ to $z = 6$, with 48 analysis snapshots between $z = 15$ and 6. Primordial hydrogen/helium cooling and a star-formation prescription are included. The cohort consists of the gas particles in a host at first entry. Progenitor labels come from group membership one dynamical time earlier; ungrouped gas is divided into spatial octants. Several partitions of the current host measure the remaining label information. [C2]

Approximately half of the resolved label information is already lost at entry. The measured kernels are fitted by

$$K_{\rm mix}(\tau) = 1 - (1 - K_0)e^{-\tau/\tau_{\rm mix}}, \qquad K_0 \simeq 0.5. \tag{76}$$

The subsequent decay takes a few dynamical times. Because this kernel includes an immediate component, its fitted tail time cannot be directly equated with the pure-exponential f in the cosmological fit. The appropriate comparison convolves each measured kernel with its corresponding first-entry history.

Retention gate	Hosts	$I_{\rm rel}$ across partitions	Reference target
$17\,\mathrm{km\,s^{-1}}$	3	75.4–75.9	94.2
$14\,\mathrm{km\,s^{-1}}$	10	93.1–96.7	94.2
$12\,\mathrm{km\,s^{-1}}$	16	108.0–109.9	94.2

Table 2: The measured mixing histories after convolution with assembly. Ranges show changes among spatial partitions, not statistical confidence intervals. Retention gates were imposed in the analysis.

At the reference $17\,\mathrm{km\,s^{-1}}$ gate, the measured integral is about 20% too small. A $14\,\mathrm{km\,s^{-1}}$ gate brings it into the required range. That agreement identifies a physical possibility, but selecting a gate after examining the integral does not independently predict its normalization. At the reference gate, changing the lookback time of the labels moves the integral to 82–83; using only grouped gas moves it to 57–65. The definition of the retained source record is therefore consequential.

The run resolves only about 500–2400 gas particles per host, includes three hosts at the reference gate, and omits molecular cooling, feedback, and explicit subgrid diffusion. Its spatial partitions scale with the evolving halo radius. A physical prediction requires label and resolution rules fixed by the theory, gas-retention thresholds measured from the simulation, and convergence under improved gas resolution and transport physics. The present run supports a measurable assembly-and-mixing mechanism while leaving its absolute release rate unsettled.

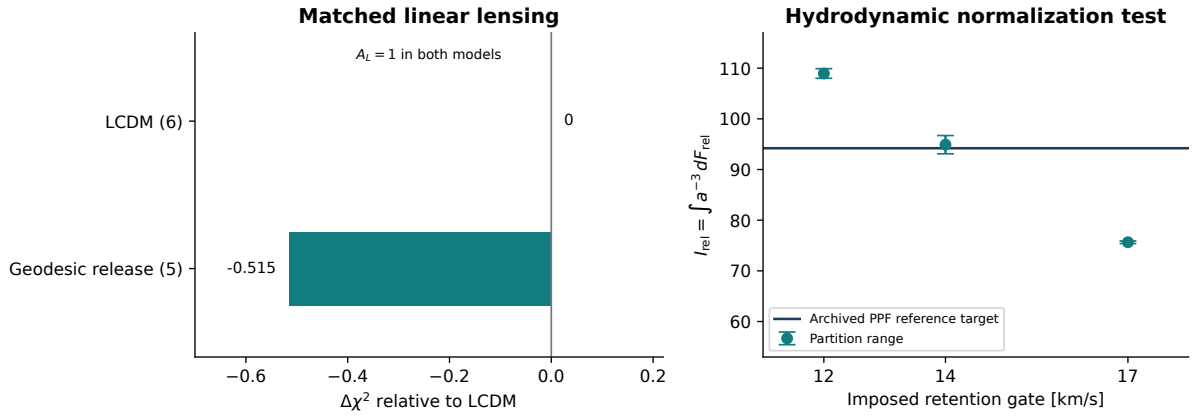


Figure 6: Left: the matched linear-lensing likelihood comparison, with cosmological parameter counts in parentheses. Right: the archived hydrodynamic kernels compared with the linear-depletion PPF reference target $I_{\rm rel} = 94.2$. The ranges describe partition choices; the positive-allocation model uses a different weighted integral.

8.9 The vacuum density from the retention gate

Equation (66) turns the release integral into a vacuum density. With $\omega_b = 0.02238$ and the tied abundance, $I_{\rm rel}$ alone fixes h and Ω_Λ. The integral weights each released cohort by the matter density at its release, a^{-3}, so the earliest retaining halos dominate it. Their number is set by the retention gate, the threshold above which a halo keeps cooling gas in which progenitor labels mix. A lower gate admits more halos at earlier times and raises the integral.

The construction places records in condensed baryons: stars and gas that has cooled. Shock-heated gas whose cooling time exceeds its dynamical time carries none. A newly assembled halo therefore enters the release history when its virialized gas can cool within one dynamical time. This is the atomic-cooling limit of first-galaxy formation. For primordial gas in collisional ionization equilibrium, with atomic hydrogen and helium cooling [16], Compton cooling on the microwave background, and no molecular hydrogen or ultraviolet background, the condition $t_{\rm cool} = R_{\rm vir}/V_{\rm vir}$ at the mean virial gas density gives $T_{\rm gate} = (0.99\text{–}1.07) \times 10^4\,{\rm K}$ for $30 \geq z \geq 8$. Collisional excitation of hydrogen sets this temperature. Its rate rises by three orders of magnitude between 8×10^3 and $1.5 \times 10^4\,{\rm K}$, so the threshold barely moves with density or redshift. The virial relation $T_{\rm vir} = \mu m_p V_{\rm vir}^2/2k_B$ converts it to a circular velocity. Gas at the threshold is nearly neutral, $\mu = 1.22$, which gives $V_{\rm gate} = 11.6\text{–}12.0\,{\rm km\,s^{-1}}$. The reference gate of $17\,{\rm km\,s^{-1}}$ is the same temperature evaluated for fully ionized gas, $\mu = 0.59$.

The archived run measures a condensation threshold. Its group catalogues record the star-forming gas and stellar mass of every group at each snapshot. The fraction of groups holding condensed baryons rises from below 0.1 to unity across 0.03 dex in virial velocity and reaches one half at 8.6, 8.9, 9.3 and $10.2\,{\rm km\,s^{-1}}$ for $z = 11\text{–}15$, 9–11, 7.5–9 and 6–7.5. Halo centres are denser than the virial mean, so gas condenses in smaller halos than the mean-density estimate allows. Using this threshold as the release gate assumes that condensed baryons identify recoverable source records.

Table 3 recomputes the first-entry history at each pre-reionization gate and convolves it with the measured kernels. The post-reionization gate stays at $30\,{\rm km\,s^{-1}}$ and reionization at $z_{\rm re} = 7.66$. The six kernels measured on the ten hosts at $14\,{\rm km\,s^{-1}}$ and the sixteen hosts at $12\,{\rm km\,s^{-1}}$ change $I_{\rm rel}$ by at most 6% at fixed gate, so the gate controls the result. The three hosts at $17\,{\rm km\,s^{-1}}$ have a slower tail and give the integral 75.4–75.9 listed above.

Origin of the pre-reionization gate	$V_{\rm gate}$ (${\rm km\,s^{-1}}$)	$I_{\rm rel}$	h	Ω_Λ
Run, condensed baryons, $z = 9\text{–}15$	9	128.6–135.7	0.77–0.79	0.766–0.775
Run, condensed baryons, $z = 6\text{–}9$	10	118.9–125.4	0.75–0.77	0.752–0.761
Cooling criterion, neutral gas	12	103.7–109.5	0.71–0.73	0.725–0.736
Gate selected by the likelihood	14	92.4–97.5	0.69–0.70	0.702–0.713
Ionized conversion of cooling temperature	17	80.9–85.3	0.65–0.67	0.673–0.685

Table 3: Release integral, Hubble parameter and vacuum fraction from Equation (66) against the pre-reionization retention gate. Ranges span the six kernels measured at the 12 and $14\,{\rm km\,s^{-1}}$ gates. The archived linear-depletion likelihood requires $I_{\rm rel} = 94.2 \pm 0.8$, $h = 0.689$ and $\Omega_\Lambda = 0.707$.

Both gates computed without feedback lie below the gate that the likelihood selects. The cooling criterion gives $\Omega_\Lambda = 0.73$ and the run gives 0.75–0.77. Supernova feedback and photoheating inside ionized regions before the global reionization redshift remove gas from the smallest halos and raise the effective gate [17, 18]. The run contains neither process. Gates from 12 to $17\,{\rm km\,s^{-1}}$ span the neutral and ionized conversions of the cooling temperature. Across that range $\Omega_\Lambda = 0.67\text{–}0.74$, and the vacuum density lies between 14% below and 16% above the

likelihood value.

The cooling-criterion calculation uses η_*, the formation abundance, assembly of the measured fluctuation spectrum, the measured mixing kernel and the atomic-hydrogen cooling threshold. It gives $I_{\rm rel} = 103.7$–109.5, 10–16% above the archived target 94.2, and $\Omega_\Lambda = 0.725$–0.736 against 0.707. Condensation in the no-feedback run gives $I_{\rm rel} = 118.9$–135.7. Neither calculation includes feedback. The likelihood selects a gate of $14\,{\rm km\,s^{-1}}$, about $5 \times 10^7 M_\odot$ at $z \simeq 10$.

A retained well is a halo that keeps gas able to cool, in which progenitor labels mix. The post-reionization gate measures this retention for photoheated gas [18], and the same criterion holds before reionization. Stars carry records and are collisionless, so they supply no label mixing; a halo that forms a few stars and then loses its gas lies outside the gate. The fraction of halos holding star-forming gas therefore measures the pre-reionization gate. The Renaissance simulations follow Population III and metal-enriched star formation with radiative transfer, radiative and supernova feedback, metal cooling and Lyman–Werner backgrounds to $z = 15$, 12.5 and 8 in rare-peak, normal and void regions [19, 20]. About 10% of halos of $10^7 M_\odot$ host star formation younger than 20 Myr, about half of those of $10^8 M_\odot$ do, and all above $10^{8.5} M_\odot$ do, nearly independently of redshift and region. Treating this occupation as the probability that a trajectory's gate lies below a given mass, and using the linearity of the release integral in the first-entry history, Table 4 gives the result. No input in the chain is adjusted to the vacuum density.

Pre-reionization gate	$I_{\rm rel}$	h	Ω_Λ
Sharp, $10^7 M_\odot$	137–145	0.81	0.78
Sharp, $10^{7.5} M_\odot$	104–109	0.72	0.73
Sharp, $10^8 M_\odot$	80–84	0.66	0.68
Renaissance occupation	92.3–97.4	0.693	0.709

Table 4: Release integral, Hubble parameter and vacuum fraction with a mass gate before reionization. The post-reionization gate is $30\,{\rm km\,s^{-1}}$ and $z_{\rm re} = 7.66$. Ranges span the six measured kernels. The likelihood requires $I_{\rm rel} = 94.2 \pm 0.8$, $h = 0.689$ and $\Omega_\Lambda = 0.707$.

The kernels move the occupation result by ± 2.5 in $I_{\rm rel}$, the shape of the occupation curve by $+4/-2.4$, the reionization redshift $z_{\rm re} = 6.94$–8.40 by ± 5.5, and a post-reionization gate of 25–$35\,{\rm km\,s^{-1}}$ by ± 4.3. In quadrature,

$$I_{\rm rel} = 95.5 \pm 8, \qquad \Omega_\Lambda = 0.709^{+0.017}_{-0.019}, \qquad h = 0.693 \pm 0.021, \tag{77}$$

against the required 94.2, 0.707 and 0.689. Every term of the flatness sum is fixed once the baryon density is measured: the tie gives the committed density $5.333\,\omega_b$, the release gives $\omega_\Lambda = \eta_* \omega_c I_{\rm rel}$, and flatness gives h^2 as their sum. The boxed h is therefore a prediction, $H_0 = 69.3 \pm 2.1\,{\rm km\,s^{-1}\,Mpc^{-1}}$, with no fitted dark-sector parameter, and h varies nearly as $\omega_b^{1/2}$. The reionization redshift and the post-reionization gate dominate the error. The Milky Way satellite census places the half-occupation peak mass for any star formation near $10^{7.5} M_\odot$ [21]; those faint galaxies form a few thousand solar masses of stars and lose their gas, so they lie below the retention gate. For comparison, the standard quantum estimate of the vacuum energy exceeds the measured value by about 10^{120}, and ΛCDM fits Ω_Λ as a free constant. A measured cold-gas retention fraction at $z \geq 8$, a tighter reionization redshift and a feedback-resolved mixing kernel sharpen the test; a converged integral that differs from the target by more than its uncertainty falsifies the normalization.

The occupation gate, the measured kernels and the post-reionization gate fix the release history once the reionization redshift is given, and the optical depth measures that redshift. Taking

$z_{\rm re}$ from the fitted τ removes the mixing time, and the release model has four cosmological parameters against six for ΛCDM. In the matched likelihood with $\sum m_\nu = 0.06\,\mathrm{eV}$, nonlinear lensing and free calibration the frozen history gives $\chi^2 = 2240.100$ against 2239.031 for ΛCDM, $\Delta\chi^2 = +1.07$, at $z_{\rm re} = 8.12$, $\tau = 0.0587$, $I_{\rm rel} = 92.0$ and $H_0 = 68.42$. With massless neutrinos, fixed calibration and the low-redshift offset it gives 2226.483 against 2226.767. The extreme kernels and post-reionization gates of 25–35 $\mathrm{km\,s^{-1}}$ give $\Delta\chi^2 = +1.1$ to $+2.6$. Holding $z_{\rm re}$ at 7.66 independently of τ gives $\Delta\chi^2 = +20.2$ and $+4.0$, so the frozen history predicts that a larger optical depth accompanies a smaller vacuum density. With $a_0 = \epsilon c H_0$ the first fit gives $a_0 = 1.249 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, and the astrophysical chain without cosmological data gives $(1.27 \pm 0.04) \times 10^{-10}\,\mathrm{m\,s^{-2}}$, against $1.20 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ from rotation curves [30]. Neither uses galactic data. [C2, C7, C8]

8.10 Equation of state, vacuum energy, and decisive tests

The vacuum stress has intrinsic $w_\Lambda = p_\Lambda/\rho_\Lambda = -1$. Its density grows because it receives energy. If one describes this growing density by a fictitious separately conserved fluid, its inferred equation of state is

$$w_{\rm eff} = -1 - \frac{1}{3}\frac{d\ln\rho_\Lambda}{d\ln a} = -1 - \frac{Q}{3H\rho_\Lambda} \le -1. \tag{78}$$

This effective phantom sign comes from energy transfer. A reconstructed decrease of the physical vacuum density would contradict the monotone release law, provided the matter transfer and observational systematics were modeled consistently. A fitted w_0, w_a curve under conserved-CDM assumptions is a different inference problem; its crossing of -1 cannot alone decide this interacting model.

Normalized microscopic probabilities are unchanged by a common shift of their energy origin. Coupling to gravity also requires the absolute stress-energy assigned to that origin. The current cosmological construction selects zero initial vacuum density and generates the subsequent density through release. Establishing that selection from the microscopic gravitational theory remains necessary to claim a solution of the absolute vacuum-energy problem. The empirical fit tests the generated component under the selected initial condition.

The saved exponential-allocation calculation evolves the geodesic-clock perturbations with zero rest-frame sound speed and $A_L = 1$. Nonlinear lensing corrections are disabled in both models, and the late-radiation truncation that introduced reference-split dependence is disabled. The assembly table retains its reference power spectrum while cosmological parameters vary. Both models use massless neutrinos with $N_{\rm eff} = 3.044$ and the same simplified low-ℓ treatment. [C4, C5]

A full test requires a validated nonlinear response, assembly recomputed for the sampled cosmology, the full low-ℓ likelihood, and an independently measured release kernel. The selected depletion completion below inherits the cold linear fit; its nonlinear spatial predictions require a separate test. The present result demonstrates near-equal fit quality with a microscopically tied abundance and a physically interpretable release parameter in the specified pipeline.

8.11 A self-limiting local completion and its coupling

The unit-clock action is exact above a selected depletion threshold. For $X = -\nabla_\mu\phi\nabla^\mu\phi/2$, a covariant continuation is

$$P = \lambda X - e - V, \quad L = 3(K - V) > 0, \quad e = \frac{\lambda}{2} + \frac{2[L-\lambda]_+^3}{3L^2}, \quad \lambda \ge L/2. \tag{79}$$

The constant K is a minimum total energy. For $\lambda > L$ the Lagrangian is exactly $\lambda(X - 1/2) - V$. On the intermediate branch $r = \lambda/L \in (1/2, 1)$, define $v = \sqrt{2X}$ and $j = \lambda v$. Then $X =$

$1/2 - 2(1-r)^2$ and

$$\dot{j} = -\left[\theta + \frac{V_\phi A(r) j}{\lambda^2}\right] j, \qquad A(r) = \frac{1 + 2r - 2r^2}{3 - 2r} \in [3/4, 1]. \tag{80}$$

For regular flow with $K - V$ bounded away from zero, the current stays nonnegative. Its depleted homogeneous endpoint has $j = 0$, $\rho = K$, $p = -K$. The regular sound speed lies between zero and one. The constant minimum removes the scalar force at rest. This slows the source field relative to proper time; no universal atomic-clock effect has been established. The cubic and floor are constitutive choices. [D1]

For the fitted history, the choice $K = V(1) + B[e^{-\eta_* F(1)} - e^{-\eta_*}]$ bounds all remaining future release under the stated monotonicity assumptions. The saved value is $K = 0.3369964$ in units $3H_{100}^2/(8\pi G)$, with $\max_{a \le 1} L/\bar{C} = 0.0299937$. The background and a neighborhood of its linear perturbations stay on the original branch. Hence the completion inherits the fitted cold adiabatic linear cosmology. Deep voids enter a pressure regime with a changed minimum energy and require new expansion, profile, and lensing predictions. Nonzero-null-gradient interfaces and nonlinear caustics remain unresolved.

A finite-work contact extends this mechanism to redistribution. For a transition $E \to E'$ toward target h, let $Q = \Theta \ln(h'/h)$, $W = E' - E + Q$, $U' = U - W$, and $E'_B = E_B + Q$. Their total energy is conserved. A symmetric gate $v(U)v(U - W)$ allows only admissible endpoints and vanishes at the floor. Central pair impulses give reciprocal momentum exchange. The target and physical rate require selection beyond detailed balance. [D2]

A warm fluid stress stores heat as $e_T = zn^\gamma \ge 0$ with $1 < \gamma \le 2$ and $\rho = R_0(n, \phi) + e_T$, $p = p_0 + (\gamma - 1)e_T$. Its sound speed is an enthalpy-weighted average of the cold speed squared and $\gamma - 1$, preserving the causal local bound and energy floor. The gate $g = n\mu/(\lambda_0 + \gamma e_T) \in [0, 1]$ retains a regular signed power gB_c and reduces to the cold v_0^2 gate. In the selected $\gamma = 4/3$ compression test, five resolutions up to 2048 cells conserve mean total energy to 4.9×10^{-15} and keep positive current. Generated heat falls from 0.0641 to 0.0140 under refinement, so physical dissipation has not converged. A separate particle-feedback test has fractional energy error 1.02×10^{-5} and stops short of the target when available work is insufficient. These tests establish a conservative effective implementation; the coupled spatial and multistream evolution remains to be validated. [D2]

9 Clusters: conservative dynamics and observed targets

The cluster calculation uses positive, inertial stock, gas, and stars in one gravitational potential. Gas pressure and shocks can separate the gas from the weakly coupled stock and galaxies during an encounter. The observed lensing map measures their total projected mass. This construction supplies a transport mechanism and quantitative targets; the present comparisons do not constitute a fitted cluster population. [CL1–CL3]

9.1 A common mass budget and a conservative encounter

For a fully released closed cohort, the adopted normalization in the formation mass convention of the cosmological fit is

$$k_C = 5.333\, e^{-\eta_*} = 5.1760752, \qquad C_0 = k_C B_0. \tag{81}$$

The bare convention gives $e^{-\eta_*}/\epsilon = 5.164$. If its baryons and stock remain coextensive, the baryon fraction is $1/(1 + k_C) = 0.1619151$. More generally, an aperture retains different fractions

of the ancestral components. The measured target can be expressed as

$$C_{\mathrm{req}} = M_{\mathrm{grav}} - B, \qquad \eta_{\mathrm{ret}} = \frac{k_C B}{C_{\mathrm{req}}} = \frac{f_b^{\mathrm{ret}}}{f_C^{\mathrm{ret}}}. \tag{82}$$

This distinguishes the common cohort normalization from the distribution produced by assembly. A positive inventory exists for $C_{\mathrm{req}} > 0$: $B_0 = \max(B, C_{\mathrm{req}}/k_C)$ leaves nonnegative exterior baryons $B_0 - B$ and stock $k_C B_0 - C_{\mathrm{req}}$. It does not determine a realizable transport history or its energy cost.

The encounter model uses central pair impulses with opposite momentum changes. If w is the radial relative velocity and μ the reduced mass, the kinetic change is $\Delta K = \mu(w'^2 - w^2)/2$. Explicit work and heat registers balance that change, and a symmetric gate suppresses transitions with an endpoint at the energy floor. Gas and stars from either progenitor can be local partners. The target distribution, rate, range, and initial work reserves are selected effective inputs. [D2, CL1]

Fourteen encounters and three isolated controls test a common Plummer mass ratio of 0.3, with 1344 particles at baseline and 2688 in the resolution control. Weak contact leaves stock near the galaxies while gas lags. A floor-state run permits no events and reproduces the no-contact evolution; a low-work run permits 61 events against 7114 with the larger reserve. The largest energy drift is below 0.007% of the binding energy; pair momentum conservation holds at roundoff. The strongest tested contact changes the isolated stock median radius by -27% and gas median radius by $+46\%$ in six dynamical time units. That control prevents interpreting a higher contact rate as an established cluster solution. The particle encounters test the finite-work coupling; they do not evolve the full spatial stress of the depleted scalar.

9.2 X-COP profiles and independent weak-lensing masses

The public X-COP archive supplies twelve hydrostatic radial profiles, seven with stellar mass data. Five overlap the MENeaCS lensing catalogue. The comparison adopts the observational distance convention $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_m = 0.3$. Projected stellar masses are approximately deprojected using the stated profile model. Unmeasured intracluster light, hydrostatic departures, triaxiality, and profile covariance remain relevant. [34, 35][CL2]

All nominal NFW hydrostatic targets admit positive cumulative stock and positive stock in fifteen tested shells. At the hydrostatic R_{500}, the coextensive-to-hydrostatic total-mass ratios for A1795, A2029, A2142, A2319, A644, A85, and ZW1215 are respectively 0.903, 0.912, 1.023, 1.213, 0.851, 0.952, and 0.679. These reconstructed totals are inputs to a target audit, rather than stock profiles predicted from the observed baryons.

Cluster	Coextensive M_{500}	Lensing M_{500}	η_{ret}
A1795	3.96–4.23	9.3 ± 2.2	0.518
A2029	7.46–8.07	12.1 ± 2.5	0.704
A2142	9.29–10.02	9.7 ± 2.3	0.969
A85	5.20–5.43	5.7 ± 2.2	0.937
ZW1215	4.19–4.46	3.5 ± 2.2	1.143

Table 5: Masses in $10^{14} M_\odot$. The coextensive prescription solves its own overdensity radius; its range doubles the nominal stellar mass and is a sensitivity interval. Retention ratios use the nominal baryons within each central lensing target's radius. Lensing errors are the published NFW marginal errors. No combined exclusion significance is assigned.

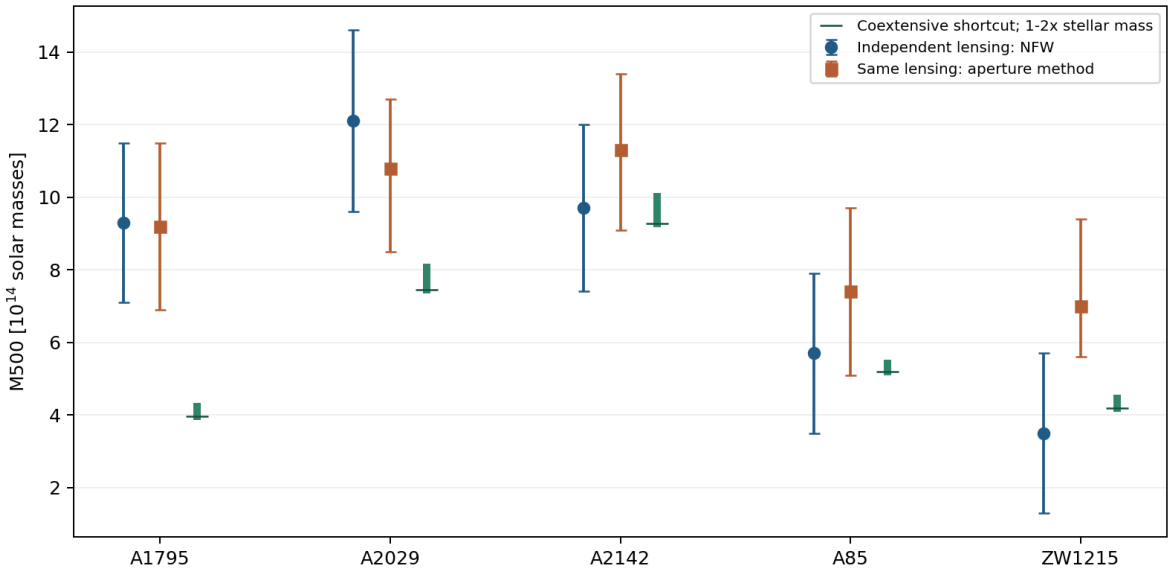


Figure 7: The common coextensive prescription and two mass estimators from the same lensing data. Their errors are correlated. A1795 and A2029 require a transport or reconstruction explanation; A2142 and A85 are useful normalization controls. ZW1215 is sensitive to the estimator.

9.3 Why clusters remain fair samples

The stock that reaches the galactic response law in isolated galaxies (Section 3.7) is the same stock that fills clusters. The selection acts through the cells of active carriers, on the stock each carrier carries. A carrier's records are identified with its condensed baryons, stars and cold gas; virialized hot gas relaxes toward the Gibbs state of its potential. A carrier's support ends where its records stop dominating the enclosed baryons, and its cells act on the stock bound to the potential of its records and the stock it carries. In the X-COP clusters the supports and cell energies are:

Cluster	r_d (kpc)	r_d/R_{500}	condensed fraction at R_{500}	cell energy ((km/s)^2)
A85	47	0.038	0.025	21
A1795	65	0.056	0.044	90
A2029	76	0.054	0.052	93
A2142	68	0.048	0.047	37
A2319	99	0.072	0.035	121
A644	87	0.069	0.051	132
ZW1215	97	0.071	0.038	55

Here r_d is the radius where projected stellar mass equals gas mass, an upper bound, and the cell energy is per unit cluster stock. Outside r_d, and for stock crossing it unbound, the cluster stock is collisionless and keeps the cohort ratio of the common mass budget. The X-COP medians test that ratio:

	total/baryon	source
Cohort prediction $1 + k_C$	6.18	theory
X-COP median, R_{500}	6.45	hydrostatic, $f_\star = 0.015$
X-COP median, R_{200}	6.10	hydrostatic, $f_\star = 0.015$
Response law at R_{500} ($g_{\rm bar} = 0.05$–$0.07\,a_0$)	4.2–5.0	theory
Cohort stock plus an always-on response	9.4–10.2	theory

A response acting on cluster stock fails in either direction. Galaxy outskirts that obey the law lie at the same $g_{\rm bar}/a_0$, at 35–50 kpc in a Milky Way model, so the switch cannot depend on acceleration alone.

Contact runs start from the observed profiles of the seven clusters with the archived contact kernel, conservative softened gravity and fixed baryons, and compare with matched no-contact controls after 8–10 Gyr, at three contact proposals per orbit:

Rule	R_{500}	$0.3R_{500}$	$0.1R_{500}$
Cells throughout, galactic work allowance	0.547–0.787	0.319–0.440	0.209–0.366
Support only, galactic work allowance	0.976–0.997	0.867–0.961	0.369–0.731
Support only, cell work	0.997–1.007	0.990–1.019	0.563–0.937
Support, carried stock, cell work	0.9998–1.0008	0.997–1.002	0.985–1.007

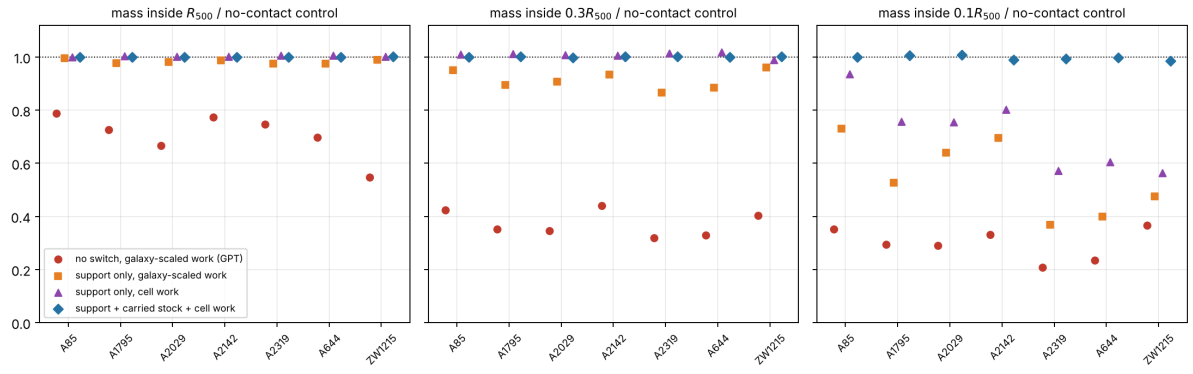


Figure 8: Mass inside three radii relative to the no-contact control for the four rules of the table, with 2000, 2000, 10^4 and 1.6×10^4 particles.

Contacts are confined to 4–7% of R_{500} with 10^{-4} of the binding energy, so the outer result follows from the support. With support only, the central galaxy's cells drain the uncarried cluster stock in its support to 7–30% of the control. With carried stock, averaged over 8–10 Gyr, the core keeps 94–105% inside r_d and 99–101% inside $2r_d$, with a Poisson error of 6–14%. The central galaxy's carried stock falls from 1.1–4.3 to 0.57–0.85 times its galactic-law value, and its support keeps 7–11 times that value in uncarried stock. The same rule leaves isolated galaxies on the law inside their supports, which for the Milky Way end at 115–490 kpc. The X-ray ellipticals on the radial acceleration relation [30] test the record identification directly. Over the radii used there, stars dominate their enclosed baryons; in NGC 720 hot gas becomes the dominant baryonic component only outside about 200 kpc [31]. Excluding the hot gas from the record source lowers the outermost points by at most 0.04 dex. Isolated ellipticals hold no group stock and should follow the law out to the radius where their hot gas dominates. Carrier support makes four observational distinctions:

- central galaxies of groups and brightest cluster galaxies depart from the rotating-galaxy relation, with an acceleration scale about 17 times larger for brightest cluster galaxies [26, 27];
- X-ray groups at 0.5–2 keV require mass beyond the response law [28], as retained collisionless stock supplies;
- in X-ray groups the excess over the law begins in the core: in A262, MKW4 and RXJ1159.8+5531 it exceeds 0.1 dex within 5–13 kpc, while hot gas overtakes the condensed baryons only at 85–210 kpc [45, 46], so the excess traces the uncarried group stock;
- isolated ellipticals follow the law out to the radius where their hot gas outweighs their stars; NGC 720 lies within about 0.1 dex of it at 102 kpc in mass models extrapolated beyond the X-ray data [31].

Resolved member galaxies and cosmological assembly are the open parts. [G8, G9, G10, G11]

9.4 What the outliers require

The central A1795 and A2029 lensing targets require respectively 48.2% and 29.6% of the minimal cohort's initial baryons outside the aperture; for A1795 this is $7.87 \times 10^{13} M_{\odot}$. Twelve finite transport maps preserve positive marginals for the seven hydrostatic and five lensing targets. This inventory construction leaves the assembly time and work requirement undetermined. [CL3]

The gas must obey the same total potential,

$$a_g = -g - \frac{P'_{\rm th} + P'_{\rm nt}}{\rho_g} + \frac{f_{Cg}}{\rho_g}. \tag{83}$$

Segregation, acceleration, and reciprocal contact are possible contributions. Passive drag vanishes at zero drift, so stationary contact support requires its own physical law. XRISM's low turbulent support in A1795 and A2029 constrains a purely turbulent explanation. [37, 38] The conditional pressure audit also depends on inner lensing extrapolation and thermal reconstruction; a joint fit remains open.

A644's 58-knot forward profile contains 11 negative total-mass increments and 15 negative stock increments. A monotone reconstruction requires 2.156 marginal error units of maximum displacement; absent joint covariance, this is not an exclusion significance. Positive parametric profiles require comparison with the underlying gas data. For ZW1215, at 1320 kpc the correlated NFW-lensing, aperture-lensing, and hydrostatic estimates are 4.32, 7.0, and $7.41 \times 10^{14} M_{\odot}$. Both systems retain reconstruction uncertainty rather than a new gravity coefficient.

9.5 The Bullet Cluster supplies a spatial target

The CANUCS Bullet v1 best-fit maps separate a total lens reconstruction from a supplied gas component. At lens redshift 0.296 and $D_{ls}/D_s = 1$, their normalization gives $\Sigma_{\rm crit} = 1.8268761 \times 10^9 M_{\odot}\,{\rm kpc}^{-2}$ and $1'' = 4.41302$ kpc. Registered apertures around BCG3 give

Radius	$M_{\rm total}/10^{13} M_{\odot}$	$M_g/10^{13} M_{\odot}$	Gas fraction
60 kpc	2.151	0.099	0.046
150 kpc	8.511	0.618	0.073

The released subcluster halo centre lies 3.92 kpc from BCG3. The supplied gas-density map peaks about 169 kpc away within the specified search region, with an 8.68 kpc gas pixel. This is a density-map diagnostic, rather than an X-ray brightness peak or an offset posterior. Subtracting gas leaves a component that still includes stars. [36][CL2]

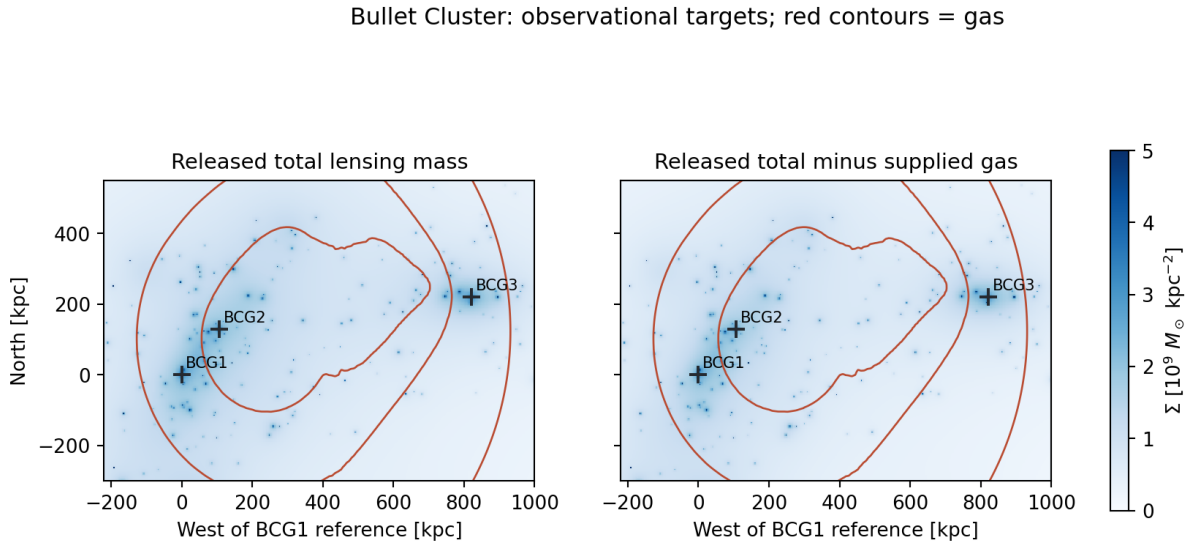


Figure 9: Observed targets from the released Bullet v1 lens model. The red contours show its supplied gas component. Both panels are derived from the public reconstruction; no simulated map is fitted here.

Aperture masses change by at most 0.214% under pixel coarsening; this is a numerical check and excludes observational uncertainty. Under an illustrative $10^{15} M_{\odot}$, 300 kpc scaling, the encounter calculations have 24–30 kpc force softening and roughly 90 kpc map smoothing. They cannot establish agreement with the few-kiloparsec halo–galaxy alignment. A successful forward test must preserve isolated cluster profiles and then reproduce lensing, gas morphology, and offsets using one selected contact law and independently specified assembly conditions.

10 Joint interpretation

The empirical contribution is a common prescription that connects a galactic acceleration scale, a retained mass distribution, tidal evolution, and cosmological energy transfer. The calculations use the same microscopic entropy and conversion fraction in several settings. Their success would be more constraining than unrelated curve fits because changing those coefficients would move several observables at once.

The LMC enclosed masses and NGC 628 vertical dispersions support the retained-stock prescription against the tested external-field and algebraic QUMOND comparisons. Crater II gives a favorable comparison with a specified NFW population, with strong dependence on the error budget. The reference acceleration-line projection agrees, while the high-redshift catalogue test retains a residual trend. The matched cosmology recovers CMB and distance data with a tied abundance; ACT lensing and E_G add compatible checks. These results establish the scope of the particular comparisons summarized in Table 1. Antlia 2, cluster outliers, Galactic mass, and independent release normalization remain active constraints.

10.1 The derivation has identifiable physical commitments

This chain explains why the empirical results bear on the microstructure: the model does not choose a separate acceleration coefficient for the LMC and the redshift sample, or a separate release fraction for each cosmological data combination. It also identifies which steps the data cannot determine by themselves. A fit to an effective gravitational field does not uniquely identify the microscopic state space that produced it.

Step	Result supplied	Physical commitment or outstanding test
Finite face and cell ensemble	$16 = 7 + 9$; 1680 cell states; $g = 7.4198000$ and $\eta_* = 0.02986684$.	The spin, matching vertex, occupancy rule, closure invariant, and evidence measure define the ensemble.
Horizon and transverse response	$\epsilon = g/(2\pi)^2$; $a_0 = \epsilon cH$; Bose response.	The tilt-mode lever $\ell = g\,c/H$ in the horizon bath connects the count to acceleration and fixes the mode energy.
Retained galactic stock	An inertial positive mass component, inherited by carriers and transported by collisionless evolution.	Assembly budgets, carrier assignment, profile preparation, and relaxation must be specified independently of the target observables.
Cosmological conversion	$\eta = \eta_*$ and an initial abundance near ϵ^{-1}.	Common energy conversion, the selected pre-acoustic formation map (whose microscopic vertex remains open), and the absolute vacuum matching condition enter the physical identification.
Assembly and mixing	A cumulative release function and a weighted vacuum integral.	Retention gates, progenitor labels, and transport set the normalization; the current expansion fit determines one mixing parameter.
Geometry and perturbations	Reduced Newtonian equations and no-slip identity proved from the specified action; carrier equations proved on their regular branch.	The continuum embedding, constrained director dynamics, and the physical matching of the effective actions remain separate requirements.

Table 6: The logical connection between the microscopic model and the observables. An exact finite calculation can coexist with conditional physical identifications and an incomplete continuum derivation.

10.2 Which conventional assumptions are challenged

For the LMC, the comparison challenges the applicability of an instantaneous external-field suppression to a previously established gravitating stock. A retained distribution continues to contribute after the ambient field changes. The tested QUMOND prescriptions and the stock model make different mass predictions with the same stated baryons. Joint stream and reflex modeling can test that distinction more directly than comparing scalar enclosed-mass summaries.

For Crater II, the issue is the central structure and stripping response of the excess gravitating mass. The tested capacity profile loses enough support to produce a cold, extended remnant under the adopted orbit and initial conditions. Standard-concentration NFW halos under the tested abundance prior perform poorly in that comparison. Low-concentration and cored dark-matter alternatives require their own matched tests. Assembly history already matters in conventional dark-matter dynamics; history dependence alone cannot distinguish the two frameworks.

For the acceleration scale, the proposed change is a cosmological normalization $a_0 \propto H$. Its observed value at one epoch would cease to be an independent constant. The published intermediate-redshift relation is consistent with the fixed reference curve under the stated

projection. A direct galaxy-level likelihood would test both its curvature and its scatter.

Irreversible source-identity loss generates vacuum energy. The microscopic conversion and formation abundance tie the release history to the CMB and distance fits. The halo occupation of feedback-resolved simulations sets the retention gate with no input adjusted to the vacuum density and gives $\Omega_\Lambda = 0.709 \pm 0.02$ against the required 0.707. Gates computed without feedback give 0.73 from the hydrogen cooling criterion and up to 0.77 from condensation in the archived run (Section 8.9). The frozen history, with the reionization redshift taken from the fitted optical depth, fits the cosmological data with four parameters at $\Delta\chi^2 = +1.07$ against six-parameter ΛCDM. A measured cold-gas retention fraction sharpens this normalization test. The current evidence supports these mechanisms as testable alternatives; it does not establish the elimination of all dark-matter models, all MOND formulations, or the absolute cosmological constant problem.

10.3 A joint test of the same construction

The reference galactic normalization uses $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. The standard linear-depletion configuration gives $H_0 = 68.458$ and $a_0 = 1.25005 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, about 1.6% above the reference acceleration scale; the 22 dwarf classifications are unchanged and their χ^2 moves by 0.1. The catalogue diagnostics and old-disk check use the exponential fit's normalization, $H_0 = 68.93885586416$ and $a_0 \simeq 1.25883 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, with massless neutrinos, linear lensing and a low-redshift supernova offset. The quoted H_0 values are in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. Joint inference propagates the fitted $H(z)$ into galaxy assembly, the acceleration relation and the relaxation of existing stock.

The same relation makes galaxy dynamics an independent measurement of the Hubble constant, $H_0 = a_0/\epsilon c$. The standard linear-depletion fit's $H_0 = 68.458$ requires $a_0 = 1.250 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, and the Cepheid-calibrated 73.04 requires $1.334 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. Hubble-flow distances carry almost no information on H_0 here, because the inferred a_0 scales with the assumed distance scale in nearly the same way as the prediction. The 44 SPARC galaxies with tip-of-the-red-giant-branch, Cepheid or supernova distances, fitted with per-galaxy mass-to-light ratios, distances and inclinations under their priors, give $a_0 = (1.21 \pm 0.19) \times 10^{-10}\,\mathrm{m\,s^{-2}}$, or $H_0 = 66 \pm 11\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. All 153 galaxies of quality 1–2 give 1.03 ± 0.11, and the subsamples range from 0.97 for gas-dominated dwarfs to 1.37 for massive galaxies with absolute distances; that drift sets the present systematic floor. A 2% determination of a_0 with absolute distances separates 68.5 from 73 at 3σ. [T1, section 27.3]

The same requirement applies to the Milky Way potential. The available moving-LMC calculations use a Galactic mass within 50 kpc about 17% above the Brooks et al. inference in the stock model, and about 36% above it in the NFW model. A joint Milky Way–LMC fit would alter the satellite pericentres and the stripping histories. Those changes should be propagated to both Crater II and Antlia 2 before assigning a combined evidence ratio. [G4][6]

A decisive program follows from these dependencies:

1. Infer the Milky Way and LMC jointly using streams, halo reflex kinematics, and a common source-conserving force model. Predict additional stream observables from the resulting mass distributions.
2. Evolve the dwarf populations with resolved self-gravity and prespecified progenitor priors. Apply the actual stellar selection, velocity-gradient, and membership inference to their mock observations. Include concentration scatter and physically motivated cores in the dark-matter comparison.
3. Fit the intermediate-redshift galaxy data directly with the cosmological $H(z)$ from the same model. Preserve correlations between baryonic masses, pressure support, geometry, and

acceleration.

4. Measure retention and source-label loss in converged hydrodynamic runs with supernova feedback, photoheating and a label prescription fixed in advance, and read the retention gate from gas retention. Use that release history with the existing positive-allocation perturbation implementation and a matched full cosmological likelihood.

Each step tests an existing physical commitment of the construction.

A Provenance and numerical verification

A.1 Scope of the verification

The integrated Lean 4.35.0 library contains 160 modules and 2,937 audited theorem constants, including its legacy sublibrary. The manuscript ledger records 874 statements as proved, 90 in part, 153 as physical premises, 64 as data comparisons, and 19 as simulation outputs. Its other entries distinguish numerical calculations, definitions, and statements awaiting formalization. The saved audit reports only the standard Lean axioms and no admitted proofs. The counts include mathematical helper results. [T4]

For this paper, the relevant proofs cover the finite ensemble and closure root, the restricted freedom in the marked-register identification, positive finite allocation, and radial relaxation with preserved angular marginals. First variation with compactly supported tests gives the reduced two-potential and Newton equations. Decay at infinity gives $\Phi = \Psi$. The carrier-action proof assumes a smooth interpolating function; the adopted singular derivative at zero baryonic field limits its direct classical application. The constrained director equation, full cosmological perturbations, and observational likelihoods retain their separate analytic or numerical status.

The numerical review independently evaluates the finite ensemble, redshift projection, LMC mass profile and QUMOND flux, archived Crater II likelihoods, and a reference release history. The current verification record uses the unrounded acceleration coefficient and imports the higher-accuracy matched cosmology endpoints. The associated figures use those records. The archived release-history and hydrodynamic panels retain their original linear-depletion calibration.

The dated research records include the exponential-allocation cosmological optimization, finite-work and fluid tests, merger controls, observed cluster reductions, and outlier audits. They were not rerun to typeset this paper. Six extension modules contribute 119 explicit theorem declarations. The integrated 4.35 build, source scan, axiom audit, and ledger citation check pass; the six modules also pass standalone Lean/mathlib 4.19 checks. The existing claim ledger retains its snapshot scope. [T5] Source inspection and saved audits establish which calculation supports each statement. Numerical convergence, selection effects, and physical matching are assessed with the relevant empirical result. SHA-256 records preserve the identities of the source and result snapshots.

A.2 Local source key

Paths below are relative to the supplied `Sol and Fable` workspace. The source package includes the document, figures, numerical review, and verification script; the original simulation archives remain in that workspace.

Key	Source and use
T1	`01 Current Papers/Full/Entropic Scalar EFT.tex`. Current microscopic construction, effective galaxy dynamics, release accounting, and covariant closure; especially sections 5–6, 15, 20, N.8, and O.3–O.15. The numerical review records its source snapshot by hash.
T2	`02 Research/Spacetime consolidation 2026-10-16/MICROSTRUCTURE_TO_SPACETIME.md`. Finite geometry, interaction, matter, and clock construction.
T3	`01 Current Papers/Microstructure CDT/MICROSTRUCTURE_CDT.tex`. Selected CDT realization, normalization conventions, and finite geometry–matter–clock tests, with conventions aligned to the current Entropic Scalar EFT manuscript.
T4	Entropic-EFT-Lean, 29 October 2026 snapshot. Local library: `02 Research/Unified closure 2026-10-21/formalization`. Claim ledger, theorem sources, and saved axiom/citation audits.
G1	`04 Notes and References/Technical Notes/history_gravity_reproduction_2026-09-27.md`. Analytic galaxy checks and reproducible provenance for the stock model.
G2	`02 Research/LMC and satellites checks/qumond_flux.py` and `02 Research/LMC and satellites checks/output.txt`. External-field QUMOND surface-flux comparison.
G3	`02 Research/Crater II and Antlia 2 vs LCDM/PREREGISTRATION.md`, `PREREGISTRATION.sha256`, and `analyse_grid.py` in that folder. Core observation likelihood, priors, and hash identity.
G4	`02 Research/Crater II and Antlia 2 vs LCDM/RESULTS_with_LMC.md`, `analyse_lmc.py`, and `lmc_results.zip` in that folder. Current moving-LMC trajectories and evidence.
G5	`04 Notes and References/Technical Notes/carrier_hierarchy_calcs_2026-09-26.md`. The active-carrier inheritance calculation for tidal dwarfs.
G6	`02 Research/Crater II and Antlia 2 vs LCDM/bf_robustness.txt` and `antlia2_ji_mock/mock_approx_analysis.txt` in that folder. Model-error sensitivity and approximate observational mocks.
G7	`02 Research/Stock adjustment resolution 2026-10-21/START HERE.txt`; `02 Research/Halo replication audit 2026-10-21`. Radial adjustment, retained angular marginals, and fixed-target halo tests.
G8	`02 Research/Galaxy sector resolution 2026-10-24` and its `Repairs - 3D contacts and exact minimiser` folder. Selection constraints, exact constrained minimiser, three-dimensional contact runs, lensing bound, and independent audits.
G9	`02 Research/Contact rule physical basis 2026-10-24`. Bose relative-entropy bias, equipartition weight, zero-temperature detailed balance, Bose-functional runs, and the Lean module `Entropic/ContactSelection.lean`.
G10	`02 Research/Bose contact observed clusters 2026-10-24`. Forty contact runs on seven X-COP clusters without a cluster gate, at galaxy-scaled and fixed work allowances, with numerical audit.
G11	`02 Research/Cluster switch derivation 2026-10-24`. Record-dominated supports and cell energies of the X-COP clusters, the cell reservoir $E_{\mathrm{cell}}/M_{\mathrm{stock}} \to V_f^2$, contact runs under four rules, and gated galaxy runs.
G12	`02 Research/Highest-leverage tensions 2026-10-28`: `lensing/`, `dwarfs/`, `ELIMINATION.md` and `REVIEW_CORRECTIONS.md`. Isolated-lens population forward model with support-limited stock, and the satellite support assignment for the classical dwarfs, with independent review.

C1	`02 Research/Full Planck tied abundance/model.py`, `RESULTS.md`, and `fits_lenscorr.jsonl` in that folder; `02 Research/Dark energy prediction tests/common.py`. Matched likelihood, data handling, Boltzmann mapping, and reference ΛCDM result.
C2	`02 Research/Release zoom/RESULTS.md`, `zoom_fits.jsonl`, `zoom_fit.py`, `o12_release.py`, `Frel_table.npy`, `FZ_ref.npy`, and `kmix_summary.txt` in that folder. Archived PPF release fit, reference history, and hydrodynamic mixing test.
C7	`02 Research/Dark energy normalization 2026-10-30`: `cooling_gate.py`, `gate_from_run.py`, `irel_gate.py`, their JSON outputs, and `NOTE.md`. Retention gate from the cooling criterion and from the archived run, and the release integral, h and Ω_Λ against the gate.
C8	`02 Research/Retention gate from feedback simulations 2026-10-31`: `gate_mass.py`, `grid_mass.py`, `mixtures.py`, `renvar.py`, their JSON outputs, and `NOTE.md`. Release integral with the feedback-resolved occupation as the retention gate, and its uncertainty; `frozen_likelihood/` holds the frozen-history fits.
C3	`02 Research/Formation operator check/RESULTS.md`. Formation abundance and the 5.333 mass convention.
C4	`02 Research/High leverage closure 2026-10-21/fit_comparison.json`, `fit_physical_geodesic/result.json`, `fit_physical_lcdm/result.json`, `cosmology_checks.json`, and `growth_independent_check.json` in that folder. Direct clock-fluid implementation, matched linear-lensing fits, and numerical checks.
T5	`02 Research/Manuscript cluster and dynamics integration 2026-10-23`. Preserved source hashes, integration diffs, six-module manifest, compilation checks and PDF validation. The six modules are included in the checked extension snapshot under `Entropic/Extensions20261023`.
T6	`02 Research/Empirical fifteen-test structure 2026-10-23`. Fifteen-family coverage map, preserved source versions, editorial diffs, and validation of the two-part presentation.
U1	`02 Research/Unified empirical program 2026-10-23/execution_register.json`. Fifteen-sector evidence register; S01–S15 identify the selected datasets, saved results, source URLs, and remaining tests.
C5	`02 Research/Unified empirical program 2026-10-23/results/cosmology_final.json`. Exponential-allocation fit, matched reference, and saved observables.
C6	`02 Research/Highest-leverage tensions 2026-10-28/cosmology`: `harness_v2.py`, `run/`, and `REVIEW_CORRECTIONS.md` in the parent folder. One-harness ΛCDM and direct geodesic release fits across supernova, neutrino and lensing settings, with independent review.
D1	`02 Research/Connected theory closure 2026-10-23`. Depletion action, unchanged linear domain, lag and noise bounds, and conditional tensor calculation; 42 Lean declarations.
D2	`02 Research/Effective dynamics and coupling 2026-10-23`. Finite-work contact, warm fluid stress, compression and feedback checks; 36 Lean declarations.
CL1	`02 Research/Conservative cluster merger 2026-10-23`. Seventeen encounter/control runs, conservation and refinement records; 8 Lean declarations.
CL2	`02 Research/Observed cluster tests 2026-10-23`. X-COP and MENeaCS profiles, Bullet v1 targets, source hashes, and 15 numerical/data-handling checks.
CL3	`02 Research/Cluster outlier closure 2026-10-23`. Conditional pressure and transport audit, A644 positivity test, twelve finite transport maps, 14 saved checks, and 22 Lean declarations.

A.3 Reproduction commands

With Python, NumPy, SciPy, and Matplotlib installed, run

```
python reproduce_review.py --workspace "PATH/TO/Sol and Fable"
```

The script evaluates archived inputs without launching simulation jobs. The supplementary `refresh_verified_results.py` updates the unrounded redshift arithmetic and imports the saved exponential-allocation fit. It preserves prior outputs before writing. Its `--legacy-linear` option reproduces the earlier matched benchmark. The broader calculations and source records are identified by U1 and C5. Build the publication from the workspace root with

```
python .organization/build_pdfs.py --paper empirical
```

The build preserves previous PDFs and stores compiler files in the workspace build directory. The source archive supplies the figures for standalone typesetting.

The public dark-energy release and DES data repository are linked in the references. The current physical mixing extension is identified by its local archive and hashes; availability of the public reproduction bundle does not by itself make every subsequent calculation public.

B Notation

Symbol	Meaning
g	Cell sharing entropy in natural-log units.
$\epsilon = g/(4\pi^2)$	Tilt-mode lever coefficient over $(2\pi)^2$.
η_*	Closure evidence coefficient; also the release fraction under the common energy-conversion prescription.
a_0	Characteristic galactic acceleration scale.
g_b or g_N	Newtonian acceleration produced by the assigned baryonic source.
n_B	Occupation of the adopted transverse mode.
M_b, M_b^L	Present baryonic mass and ancestral Lagrangian baryonic mass, respectively.
C_R	Adopted retained-stock cap relative to present baryonic mass.
R_h	Projected stellar half-light radius.
$\sigma_{\rm los}$	Line-of-sight velocity dispersion, with sample and gradient treatment specified for each comparison.
Z, B	Prior-integrated likelihood and its ratio between two models.
F_Z	Fraction that has first entered a baryon-retaining well.
$F_{\rm rel}$	Cumulative fraction processed by the source-identity release kernel.
f	Mixing time in units of the formation halo dynamical time in the pure-exponential fit.
$\rho_{c0}^{\rm wb}$	Would-be present commitment density before release losses, extrapolated as dust.
$I_{\rm rel}$	Linear-benchmark release integral weighted by a^{-3}.
$I_{\eta,{\rm rel}}$	Positive-allocation integral with the additional survival factor $e^{-\eta_* F_{\rm rel}}$.
$\omega_i = \Omega_i h^2$	Physical density parameter in the cosmological fit.

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